TOWARD THE TWIN PRIME CONJECTURE: CERTIFICATES, LIMITS OF THE SIEVE, AND A WORKING PROGRAMME

JERRY KIM

ABSTRACT. This is a living working paper. It does *not* prove the twin prime conjecture, and it does not improve the best known bound on small gaps between primes. It records (i) the state of the problem as of the date above, (ii) what this project has verified for itself, by exact computation and by the Lean 4 kernel, now including the established bound $H_1 \le 246$ of Polymath8b, reproduced end to end through a certified lower bound for $M_{50,1/25}$, (iii) small results of our own: a no-go statement for a natural attempt to sharpen the pigeonhole step, on the plain and on the ε-enlarged simplex; a generating-function formula that makes certified computation with all symmetric test functions cheap; an upper bound showing that the ε-enlargement of Polymath8b cannot, by itself, reach any $k \le 29$; numerical evidence, HEURISTIC only, that at $(k, \varepsilon) = (50, 1/25)$ no dual weight depending only on t_m and $t_1 + \cdots + t_k$ does much better than ours, and heuristic evidence that the ε-enlarged simplex stops at $k = 50$; and, for the one-band hybrid supports of Stadlmann, a rigorous floor (no $k \le 28$ at his level of distribution, whatever the caps), and an exact formula for his capped functionals, which gives a certified capped ratio $kJ_T/I_T = 4.0120 > 4$ at $k = 49$, computed in exact arithmetic by two independent programs: the variational step of Stadlmann's $H_1 \le 240$ is thereby reproduced, while his reduction [19, Prop. 1] and his new equidistribution estimates, on which 240 rests, have not been verified by us; the same computation confirms the variational criterion of Charton et al. at $k = 45$ ($4.0027 > 4$) for the caps of their Table 3 but not, with our test function, for the different caps of their Lemma 9.1 (3.9737), and with those caps no polynomial in our 846-dimensional space reaches 3.99 (a ceiling certified in ball arithmetic at 3.99; it rests on the Gram weights of one program, checked exactly at one random point); optimising the test function on Stadlmann's capped support itself, with exact Gram matrices obtained by running the exact evaluator backwards, raises his margin only from 4.0120 to 4.0124; for the support of Kintali at $k = 48$ (claimed $H_1 \le 236$), our own test function of degree 22 gives $4.0061 > 4$ (two independent exact programs since version 0.12), and for the published witness of Song and Yue (same support) we recompute their capped numerator exactly: it equals their reference integer digit for digit, as does their cap-free denominator, so their criterion $48J/I_0 > 2001/2000$ holds; since version 0.11 this is confirmed by a second, independent exact program, so the variational step of $H_1 \le 236$ is reproduced (its analytic input [18, Thm. 4.18] and the reduction [18, Thm. 4.33, §§4.12–4.14] are not verified by us, and the project's bound 236 is conditional on them); version 0.12 adds an elementary observation (Bonferroni) that lets a one-sided certificate skip the most expensive inclusion–exclusion pieces (Proposition 39), and version 0.13 extends it so that no piece with three forced coordinates is needed (Proposition 40, $4.012038868 > 4$ for the 240 vector); for the next rung, 186, version 0.13 recomputes independently the two main terms I_H and $J_{\lambda,H}$ of the numerical companion of OpenAI's preprint and finds them inside its published enclosures, by two independent programs, $\rho^* J_{\lambda,H}/I_H = 1.000206087 > 500103/500000$ (its loss bound and its analytic inputs are not checked; version 0.14 recomputes its source data and cap masks exactly from their rational definitions and checks the arithmetic of its loss tables, not yet the raw loss bounds); and (iv) a programme of original directions that later versions will test. Every statement carries one of the labels LEAN-CHECKED, COMPUTED, PROVED HERE, LITERATURE, CLAIMED (a preprint not yet independently verified), or OPEN; LITERATURE (PREPRINT) marks our reading of a definition in an unrefereed preprint, COMPUTED (FLOATING POINT) a numerical value that is not a certified bound, and HEURISTIC an argument that is not a proof.

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Prepared with the assistance of Claude (Anthropic). The computations, certificates and Lean sources accompany this paper.

1. The problem and the honest position

Write p_n for the n-th prime and $H_1 = \liminf_{n\to\infty}(p_{n+1} - p_n)$. The twin prime conjecture is the statement $H_1 = 2$. It is Open.

A finite set $\mathcal{H} = \{h_1 < \cdots < h_k\}$ of integers is *admissible* if for every prime p it misses at least one residue class modulo p. Let DHL$[k, 2]$ be the statement that every admissible k-tuple has infinitely many translates $n + \mathcal{H}$ containing at least two primes, and let $H(k)$ be the least diameter $h_k - h_1$ of an admissible k-tuple. Then

$$\text{DHL}[k, 2] \implies H_1 \le H(k), \tag{1}$$

(Literature; Lean-checked for the pairs of Theorem 1), and DHL$[2, 2]$ applied to $\{0, 2\}$ is the twin prime conjecture. All progress on H_1 since 2013 consists of proving DHL$[k, 2]$ for smaller k.

Where the bound stands (14 November 2026).

$H_1 \le$	k	Source	Label
7×10^7	3.5×10^6	Zhang [21]	Literature
600	105	Maynard [10]	Literature
246	50	Polymath8b [15]	Literature
240	49*	Stadlmann [19]	Claimed†
236	48	Kintali [8]; Song–Yue [18]	Claimed‡
212	45	Charton et al. [3]	Claimed
186	40	OpenAI [13]	Claimed§
182	39	GPT-6 Astra, prompted by Sharifi [16]	Claimed
12	5	Maynard, assuming EH [10]	Literature
6	3	Polymath8b, assuming GEH [15]	Literature

* Value of k inferred by us from $H(49) = 240$ and $H(48) = 236$, not read from the source. † Its variational step is reproduced here (Corollary 27); its reduction [19, Prop. 1] and its equidistribution input are not verified by us. Song and Yue [18, Remark 4.23] point out a possible missing hypothesis in the proof of [19, Lemma 5] (a Type IIa lemma applied after swapping the factors); they say it is not a counterexample to the conclusion. In version 0.10 we read the cited step in the arXiv text (v1): in the range $\gamma \in (\frac{1}{2}, \frac{1}{2}+2\omega+\varepsilon]$ the proof of [19, Lemma 5] applies [19, Lemma 3] with the factors swapped, Lemma 3 needs its shorter factor to be Siegel–Walfisz, and neither Lemma 5 nor case (I) of [19, Definition 9] assumes this of α. So the step reads as they say. Whether the coefficient sequences that actually occur satisfy the missing hypothesis, which would repair the step, we have not checked (Remark 41). ‡ Both 236 preprints use the same support ("D182"): $k = 48$ is read from [8, 18]. Its variational step is reproduced here (Corollary 37); its reduction [18, Thm. 4.33, §§4.12–4.14] and its equidistribution input [18, Thm. 4.18] are not verified by us. An AI audit inside this project of the proof of [18, Theorem 4.18] found one repairable mis-statement (eq. (49), inherited from [14, Lemma 6.25]) and no gap (Remark 38); this is not expert review. § Its two main terms I_H, $J_{\lambda,H}$ are recomputed here and lie inside its published enclosures (Proposition 43); its loss bound and analytic inputs are not checked.

The bound 246 stood from 2014 until the last days of August 2026. It has been formalised in Lean 4, assuming only the Bombieri–Vinogradov theorem [1]. The bounds from 240 to 186 appeared within one week. They rest on Stadlmann's idea of enlarging the support of the sieve weights using equidistribution estimates for moduli with a large smooth factor. The Lean development accompanying the bound 186 is conditional on three stated inputs (two exponential-sum bounds and a family of numerical integral bounds) [13]. An independent re-run of that repository, dated 7 and 8 November 2026, reproduced the build and the numerical certificate and describes

the result as "conditionally supported, with the computation independently reproduced; not yet settled"; the two exponential-sum bounds remain axioms [12]. A further document, dated 18 September 2026 and naming an AI system as its author, claims DHL[39, 2] and hence 182; by its own abstract its Lean development verifies the last implication "under explicit numerical and finite-field premises" [16]. We had missed it in earlier versions. (These details were read by an earlier run of this project; on 12 November the file was too large for our reader and direct download was refused, so we have not re-checked them since.) As of 7 September 2026 a standard reference still listed 246 as the best fully established value [11], and the page read the same on 12 November. On the date of this version we found no report of complete independent verification of any of the 2026 claims and no claimed bound below 182. We therefore keep three numbers apart throughout: *established* (246), *claimed* (182), and *reproduced by this project*: 246 unconditionally; since version 0.6 also 240, conditionally on the reduction [19, Prop. 1] and the equidistribution estimates of [19], which we have not verified (Corollary 27); and since version 0.11 also 236, conditionally on the equidistribution theorem of [18] and their reduction [18, Thm. 4.33, §§4.12–4.14] (Corollary 37), which we have not verified either.

Why 2 is a different problem. Under the generalised Elliott–Halberstam conjecture the sieve gives $H_1 \le 6$, and Polymath8b showed that 6 cannot be improved by purely sieve-theoretic arguments: this is Selberg's parity obstruction [15]. Lowering the bound (Track A below) and proving the conjecture (Track B) are therefore different tasks. No amount of success on Track A reaches 2.

2. What this project has verified for itself

2.1. **The combinatorial half, checked by the Lean kernel.** Our Lean development is written in core Lean 4, without Mathlib. It defines primality, the twin prime conjecture, the predicate `BoundedGaps B` (infinitely many pairs of distinct primes at distance at most B), admissibility, and DHL$[k, 2]$. It contains no `sorry`, and the only axioms it uses are `propext` and `Quot.sound`. The statement DHL$[k, 2]$ is always a hypothesis, never an axiom.

Theorem 1 (Lean-checked). (1) *The twin prime conjecture is equivalent to* `BoundedGaps 2`.

(2) *If a finite Boolean check succeeds on a list $\mathcal{H}$ (it misses a class modulo every m with $2 \le m \le |\mathcal{H}|$), then $\mathcal{H}$ is admissible. The moduli $p > |\mathcal{H}|$ are handled by a formal pigeonhole argument.*

(3) *For each pair (k, D) in Table 1, for $(k, D) = (56, 278)$, $(57, 282)$, $(58, 288)$, $(59, 300)$, $(60, 304)$, $(61, 310)$, $(62, 320)$, $(63, 324)$, $(64, 330)$, and for $(102, 578)$ and $(105, 600)$, there is an explicit admissible k-tuple in $[0, D]$, and* DHL$[k, 2]$ *implies* `BoundedGaps D`. *In particular* DHL$[2, 2]$ *implies the twin prime conjecture.*

(4) DHL$[k, 2]$ *implies* DHL$[k', 2]$ *for every $k' \ge k$ (`DHL.mono`: delete shifts from the tuple). So a proof of* DHL$[j, 2]$ *with $j \le k$ gives* `BoundedGaps D` *for each listed pair (k, D).*

(5) `BoundedGaps 1` *is false (`not_boundedGaps_one`), and a pair $\{0, d\}$ is admissible if and only if d is even (`admissible_pair_iff`). So 2 is the smallest bound of this form that can hold, and the twin prime conjecture is the strongest statement of the form* `BoundedGaps D`.

Theorem 1 is elementary. Its value is that implication (1) is now a checked interface: any future proof of DHL$[k, 2]$, by anyone, yields the corresponding gap bound with nothing left to trust on the combinatorial side.

TABLE 1. Pairs (k, D) with a kernel-checked admissible k-tuple of diameter D. The tuples were found by our own search; each D agrees with the value of $H(k)$ in published tables, whose optimality we have not re-proved.

k	D	k	D	k	D	k	D	k	D	k	D
2	2	11	36	20	80	29	130	38	176	47	226
3	6	12	42	21	84	30	136	39	182	48	236
4	8	13	48	22	90	31	140	40	186	49	240
5	12	14	50	23	94	32	146	41	188	50	246
6	16	15	56	24	100	33	152	42	196	51	252
7	20	16	60	25	110	34	156	43	200	52	254
8	26	17	66	26	114	35	158	44	210	53	264
9	30	18	70	27	120	36	162	45	212	54	270
10	32	19	76	28	126	37	168	46	216	55	272

2.2. **The sieve constant, with exact certificates.** Let $\mathcal{R}_k = \{t \in [0,1]^k : \ t_1 + \cdots + t_k \le 1\}$ and, for square-integrable F supported on $\mathcal{R}_k$,

$$I_k(F) = \int_{\mathcal{R}_k} F^2, \qquad J_k^{(m)}(F) = \int \left(\int F \, dt_m \right)^2 \prod_{i \neq m} dt_i, \qquad M_k = \sup_F \frac{\sum_{m=1}^k J_k^{(m)}(F)}{I_k(F)}.$$

Maynard's theorem [10, Prop. 4.2] states that if the primes have level of distribution θ and $\theta M_k/2 > 1$ then DHL$[k, 2]$ holds. With Bombieri–Vinogradov ($\theta < 1/2$) the condition is $M_k > 4$.

We take $F = \sum_{a+2b \le d} c_{a,b} (1 - P_1)^a P_2^b$ with $P_j = \sum_i t_i^j$, as in [10, §8]. All integrals reduce to

$$\int_{\mathcal{R}_k} (1 - P_1)^A P_2^B \, dt = \frac{A!}{(k + A + 2B)!} \, G_k(B), \qquad \sum_{B \ge 0} G_k(B) \frac{x^B}{B!} = \left(\sum_{b \ge 0} \frac{(2b)!}{b!} x^b \right)^k,$$

which follows from the Dirichlet integral. The two quadratic forms are then exact rational matrices. We compute an approximate top eigenvector in floating point, round it to rationals, and evaluate the Rayleigh quotient in exact arithmetic; the last step is a rigorous lower bound whatever happened before it.

Theorem 2 (COMPUTED, exact rational arithmetic)**.** *With F a polynomial in $1 - P_1$ and P_2 only,*

$$M_{105} \ge 4.00206976, \qquad M_{102} \ge 4.00271343, \qquad M_5 \ge 2.00410541.$$

The first value reproduces Maynard's $M_{105} > 4$ with $d = 11$ (42 coefficients). The third gives DHL$[5, 2]$ and $H_1 \le 12$ under the Elliott–Halberstam conjecture, again reproducing [10]. For $d = 11, 15, 19$ the least k at which this family certifies $M_k > 4$ is $105, 102, 102$, and at $k = 101$ it gives only 3.99931 ($d = 19$): polynomials in P_1, P_2 alone stop at $k = 102$.

2.3. **All symmetric test functions.** To go further we allow every symmetric polynomial. Since $P_1, \ldots, P_k$ generate the symmetric polynomials, the functions

$$(1 - P_1)^a \, P^b := (1 - P_1)^a \prod_{j \ge 2} P_j^{b_j}, \qquad a + w(b) \le d, \quad w(b) = \sum_j j b_j,$$

span all symmetric polynomials of degree at most d (for $k \ge d$). The integrals have a closed form that we have not seen stated in this way.

Lemma 3 (PROVED HERE; possibly folklore)**.** *Let $X_1, \dots, X_k$ be independent exponential random variables of mean 1 and $E_k(b) = \mathbb{E}\prod_j P_j(X)^{b_j}$. Then*

$$\int_{\mathcal{R}_k} (1-P_1)^A P^b \, dt = \frac{A!}{(k+A+w(b))!}\, E_k(b), \qquad \sum_b E_k(b)\frac{s^b}{b!} = \left(\sum_c w(c)!\,\frac{s^c}{c!}\right)^k,$$

where $s^b = \prod_j s_j^{b_j}$ and $b! = \prod_j b_j!$. In particular each $E_k(b)$ is a positive integer, computed exactly by multiplying exponential generating functions.

Proof. For a monomial, the Dirichlet integral gives $\int_{\mathcal{R}_k}(1-P_1)^A t^\alpha\,dt = A!\,\alpha!/(k+A+|\alpha|)!$, and $\alpha! = \mathbb{E}X^\alpha$. Summing over the monomials of P^b, all of degree $w(b)$, gives the first identity. For the second, $\mathbb{E}\exp(\sum_j s_j P_j(X)) = \left(\mathbb{E}\exp(\sum_j s_j X_1^j)\right)^k$ by independence, and $\mathbb{E}\exp(\sum_j s_j X_1^j) = \sum_c \mathbb{E}X_1^{w(c)}\, s^c/c! = \sum_c w(c)!\, s^c/c!$, as formal power series. □

The J-functional reduces to the same lemma in $k-1$ variables: with $P_j = P_j' + t_k^j$,

$$\int_0^{1-P_1'} (1-P_1)^a P^b \, dt_k = \sum_{e\le b} \binom{b}{e} \frac{a!\,w(e)!}{(a+w(e)+1)!}\,(1-P_1')^{a+w(e)+1} P'^{\,b-e}.$$

So both Gram matrices are exact rational matrices. We find an approximate top eigenvector by inverse iteration with a fixed shift above $\frac{k}{k-1}\log k$, which can only converge to the top eigenvalue. We take its floating-point midpoints as exact dyadic rationals and evaluate the Rayleigh quotient exactly. The working precision of the eigenvector matters only for speed. A separate script, `code/verify_cert.py`, rebuilds both Gram matrices from scratch and re-evaluates each stored coefficient vector in exact arithmetic. (No coefficient vector is stored in the repository for the value $M_{105} \ge 4.17427$ below, so that value is not covered by this re-check.)

Theorem 4 (COMPUTED, exact rational arithmetic)**.**

$$M_{105} \ge 4.17427, \quad M_{64} \ge 4.00363, \quad M_{58} \ge 4.00927,$$
$$M_{56} \ge 4.00519, \quad M_{55} \ge 4.00516, \quad M_{54} \ge 4.00238.$$

The first two use all symmetric polynomials of degree 11 and 12 (195 and 272 coefficients). The others use the subspace spanned by $(1-P_1)^a P^b$ with b supported on even j, as in [15, §7], of degree 16, 18, 20 and 23 (307, 471, 707 and 1236 coefficients).

Basis	degree	coefficients	least k with certified $M_k > 4$
$1-P_1$, P_2	19	110	102
all symmetric	12	272	64
even power sums	16	307	58
even power sums	18	471	56
even power sums	20	707	55†
even power sums	23	1236	54†

† Only this value of k was run at this degree; smaller k were not tried.

Corollary 5 (COMPUTED and LEAN-CHECKED, conditional on LITERATURE inputs; end-to-end within this project)**.** *Assuming the Bombieri–Vinogradov theorem and [10, Prop. 4.2], both* LITERATURE*, Theorems 4 and 1 give $H_1 \le 270$.*

This reproduces a known result and is not new. Polymath8b obtained $M_{54} > 4.00238$ with even-part polynomials of degree 23 [15, Thm. 3.9], and hence $H_1 \le 270$. Our certified value in the same space, $4.002386\ldots$, agrees with theirs to every digit they print, which is an external check on our pipeline; two independent runs of it (one from an earlier session) produced the same value. Corollary 5 is the best bound certified on the plain simplex with Bombieri–Vinogradov, by Polymath8b and by us; the upper bound of the next subsection excludes only $k \le 50$ there, so $k = 51, 52, 53$ ($H(k) = 252, 254, 264$) are not ruled out by anything in this paper. Corollary 7 below takes the next rung of the ladder $600 \to 270 \to 246 \to 240 \to 212 \to 186 \to 182$.

2.4. **What the plain method cannot do.** Polymath8b proved (LITERATURE) that $M_k \le \frac{k}{k-1}\log k$ [15]. Hence (COMPUTED from that bound) $M_k > 4$ forces $k \ge 51$, and on the plain simplex DHL$[k, 2]$ requires a level of distribution $\theta > 2(k-1)/(k\log k)$ (values truncated):

k	50	45	40	35	30	20	10	5
$\theta >$	0.501	0.513	0.528	0.546	0.568	0.634	0.781	0.994

The 2026 results go below $k = 50$ by changing the support of F, not by proving a uniform level of distribution of this size.

2.5. **Beyond the plain simplex: $H_1 \le 246$.** Polymath8b's ε-enlarged simplex [15, Thm. 3.12] (LITERATURE; we re-read the statement in the arXiv text for this version) takes F supported on $(1+\varepsilon)\mathcal{R}_k$ and integrates the square of each marginal $\int F\,dt_m$ only over $(1-\varepsilon)\mathcal{R}_{k-1}$. Call the resulting constant $M_{k,\varepsilon}$. If EH$[\theta]$ holds with $1+\varepsilon < 1/\theta$ and $M_{k,\varepsilon} > 2/\theta$, then DHL$[k, 2]$ holds. Bombieri–Vinogradov gives every $\theta < 1/2$, so for every fixed $\varepsilon < 1$ the condition is $M_{k,\varepsilon} > 4$.

We extended our pipeline to this functional, following the route that [15, §7.2] describes in words. Put $L = 1+\varepsilon$, $\tau = 1-\varepsilon$ and use the basis $(L-P_1)^a P^b$ (b supported on even j). The following identities are elementary (PROVED HERE): the substitution $t = Ls$ and Lemma 3 give

$$\int_{L\mathcal{R}_k} (L-P_1)^A P^b\,dt = L^{k+A+w(b)}\,\frac{A!}{(k+A+w(b))!}\,E_k(b);$$

the inner integral over $t_m \in [0, L-P_1']$ is the formula displayed after Lemma 3 with $1-P_1'$ replaced by $L-P_1'$; and on $\tau\mathcal{R}_{k-1}$ we write $L-P_1' = 2\varepsilon + (\tau - P_1')$, so that

$$\int_{\tau\mathcal{R}_{k-1}} (L-P_1')^A P'^B = \sum_{i=0}^{A}\binom{A}{i}(2\varepsilon)^{A-i}\,\tau^{k-1+i+w(B)}\,\frac{i!}{(k-1+i+w(B))!}\,E_{k-1}(B).$$

For rational ε both Gram matrices are again exact rational matrices. We checked the code against hand computation for $k = 2, 3$ and $F \in \{1, L-P_1\}$, and against direct iterated symbolic integration (sympy, without Lemma 3 or the binomial rewrite) of every Gram entry for $(k, d, \varepsilon) = (3, 3, 1/5)$ and $(4, 4, 1/25)$: no mismatch (`code/check_eps_sympy.py`). At $\varepsilon = 0$ it is the old code path (an old certificate re-verifies unchanged). We also found that the working precision of the eigenvector search can be cut from about 6000 to 300–400 bits without changing any certified digit, which makes degree 25 feasible on a four-core machine (build 16 minutes, search 4 minutes).

Theorem 6 (COMPUTED, exact rational arithmetic)**.** *With even power sums of degree* 25 *(*1780 *coefficients) and of degree* 27 *(*2526 *coefficients),*

$$M_{50,1/25} \ge 4.0012465 \qquad \textit{and} \qquad M_{50,1/25} \ge 4.0043062.$$

The certificates are `code/cert_M50e25_D25e.json` and `code/cert_M50e25_D27e.json` (the degree-27 run took 43 minutes to build and 10 minutes to search, with about 10 GB of memory); the script `code/verify_cert.py` rebuilds both Gram matrices from scratch and re-evaluates the

stored rational vector exactly; it uses the same Gram-matrix builder, so it checks the arithmetic and the bookkeeping, while the formulas themselves are checked by the symbolic test above. Polymath8b report $M_{50,1/25} > 4.00124$ at degree 25 and > 4.0043 at degree 27 [15, §7.2]; our values agree with both to every printed digit.

Corollary 7 (COMPUTED and LEAN-CHECKED, conditional on LITERATURE inputs; end-to-end within this project)**.** *Assuming the Bombieri–Vinogradov theorem and* [15, Thm. 3.12(i)]*, both* LITERATURE*, Theorem 6 and the Lean tuple* $(50, 246)$ *of Theorem 1 give* $H_1 \le 246$*.*

This is the established bound of 2014, reproduced; it is not new. It closes the second rung of our ladder. Table 2 shows how the certified values grow with the degree and depend on ε; it is the data behind Remark 20 below.

TABLE 2. Certified lower bounds for $M_{k,\varepsilon}$ (COMPUTED, exact Rayleigh quotients, even power sums of degree D; digits truncated). Data: `runs/scan_k49.jsonl`, `runs/scan_k50.jsonl`, written by `code/scan_eps.py`; the column $k = 50$, $D = 16$ from direct runs of `code/mk_general.py`, whose output is not stored in the repository. A dash: not computed.

	$k = 49$			$k = 50$		
ε	$D = 16$	$D = 18$	$D = 20$	$D = 16$	$D = 18$	$D = 20$
0	3.88640	3.90101	3.91056	3.90173	3.91718	3.92739
1/50	3.93490	3.95065	3.96110	3.94979	3.96644	3.97761
1/30	3.93968	3.95744	3.96933	–	3.97254	3.98521
1/25	3.93674	3.95563	3.96840	3.95041	3.97034	3.98394
1/20	3.92740	3.94822	3.96243	–	3.96234	3.97744

3. RESULTS OF OUR OWN: RIGIDITY OF THE PIGEONHOLE STEP

Every bound above ends with the same step: if the weighted average number of primes among $n + h_1, \dots, n + h_k$ exceeds 1, two of them are prime, and they are at most $h_k - h_1$ apart. A natural hope is to do better with unequal weights. Fix D, and give each position a weight $c_i \ge 0$. Call a set of positions *D-separated* if any two of its shifts differ by more than D, and let $\alpha_D(c)$ be the largest total weight of a D-separated set. If no two primes among the $n + h_i$ are within D, then $\sum_i c_i \mathbf{1}_{n+h_i \text{ prime}} \le \alpha_D(c)$. Writing

$$M_k[c] = \sup_F \frac{\sum_m c_m J_k^{(m)}(F)}{I_k(F)},$$

the sieve therefore yields $H_1 \le D$ as soon as $\frac{\theta}{2} M_k[c] > \alpha_D(c)$. For $c = (1, \dots, 1)$ and $D = h_k - h_1$ this is the usual criterion. Could a cleverer c certify some $D < h_k - h_1$ that no sub-tuple certifies?

Proposition 8 (PROVED HERE; possibly folklore)**.** *Let ω be the largest number of shifts h_i contained in an interval of length D. Then for every $c \ge 0$, not identically zero,*

$$\frac{M_k[c]}{\alpha_D(c)} \le M_\omega.$$

So the weighted criterion never certifies more than the standard criterion applied to the best sub-tuple of diameter at most D.

Proof. Let G be the graph on $\{1,\dots,k\}$ joining $i \neq j$ when $|h_i - h_j| \le D$. Its stable sets are the D-separated sets and its cliques are the sets of shifts lying in an interval of length D. Normalise $\alpha_D(c) = 1$, so c lies in

$$P = \{c \ge 0 :\ c(S) \le 1 \text{ for every stable set } S\}.$$

The graph G is a unit interval graph, hence perfect, and so is its complement by the perfect graph theorem [9]. For a perfect graph the polytope cut out by the clique inequalities is the convex hull of the indicator vectors of its stable sets [5]. Applying this to the complement of G gives

$$P = \operatorname{conv}\{\mathbf{1}_K :\ K \text{ a clique of } G\}.$$

The map $c \mapsto M_k[c]$ is a supremum of linear functionals, hence convex, and therefore $M_k[c] \le \max_K M_k[\mathbf{1}_K]$. It remains to show $M_k[\mathbf{1}_K] \le M_j$ for $|K| = j$; this suffices because M_j is non-decreasing in j (adjoin a variable and concentrate F near its zero set). Write $t = (s,u)$ with s the coordinates in K and u the others, put $\sigma(u) = 1 - \sum u \ge 0$, and let $F_u = F(\cdot,u)$, a function on the scaled simplex $\sigma(u)\mathcal{R}_j$. For $m \in K$ we have $J_k^{(m)}(F) = \int J_j^{(m)}(F_u)\,du$, and $I_k(F) = \int I_j(F_u)\,du$. Rescaling F_u to $\mathcal{R}_j$ multiplies I_j by σ^j and each $J_j^{(m)}$ by σ^{j+1}, so

$$\sum_{m\in K} J_j^{(m)}(F_u) \ \le\ \sigma(u)\, M_j\, I_j(F_u) \ \le\ M_j\, I_j(F_u).$$

Integrating over u gives the claim. □

Remark 9 (Computed, consistency check for $k = 3$). Take the shifts $(0,1,2)$ and $D = 1$, so that $\omega = 2$ and $\alpha_D(c) = 1$ for both weight vectors below. With unrestricted polynomials of degree 6 on $\mathcal{R}_3$ we find the lower bounds $M_3[(1,1,1)] \ge 1.6464$, $M_3[(1,1,0)] \ge 1.3579$ and $M_3[(\tfrac12,1,\tfrac12)] \ge 1.1604$, against $M_2 = 1/(1-W(1/e)) = 1.3859\ldots$. The half-weighted functional is far below M_2, as the proposition requires.

The same rigidity on the enlarged simplex. The ε-enlarged simplex of Polymath8b [15, Thm. 3.12] takes F supported on $(1+\varepsilon)\mathcal{R}_k$ and restricts the outer integral of each $J^{(m)}$ to $(1-\varepsilon)\mathcal{R}_{k-1}$. Call the resulting constant $M_{k,\varepsilon}$. The criterion $M_{k,\varepsilon} > 2/\theta$ is valid in two cases (Literature): under EH$[\theta]$ when $1+\varepsilon < 1/\theta$, and under GEH$[\theta]$ when $\varepsilon < 1/(k-1)$. With Bombieri–Vinogradov the first case allows every $\varepsilon < 1$, and Polymath8b's $H_1 \le 246$ uses $M_{50,1/25} > 4.0043$ [15, Thm. 3.13]. Unequal weights are no more useful there.

Proposition 10 (Proved here; possibly folklore)**.** *For $0 \le \varepsilon < 1$ and every weight $c \ge 0$, $c \neq 0$, $M_{k,\varepsilon}[c]/\alpha_D(c) \le M_{\omega,\varepsilon}$, with equality for the indicator of a largest clique. Since the conditions on ε above do not get harder as k decreases, the weighted criterion never certifies more than the standard one for the best sub-tuple of diameter at most D.*

Proof. As for Proposition 8, it suffices to show $\sum_{m\in K} J_{k,\varepsilon}^{(m)}(F) \le M_{j,\varepsilon} I_k(F)$ for a clique K of size j, together with $M_{j,\varepsilon} \le M_{\omega,\varepsilon}$. Fix the coordinates u outside K, with $|u| \le 1-\varepsilon$ (other slices do not meet the J-region). The slice F_u is supported on $(1+\varepsilon-|u|)\mathcal{R}_j$, and its J-region is $(1-\varepsilon-|u|)\mathcal{R}_{j-1}$. Put $\mu = (1+\varepsilon-|u|)/(1+\varepsilon) \le 1$ and $H(x) = F_u(\mu x)$, which is supported on $(1+\varepsilon)\mathcal{R}_j$. Then $I_j(F_u) = \mu^j I_j(H)$, and each slice J-term equals μ^{j+1} times the integral of $(\int H\,dx_m)^2$ over $|x'| \le (1-\varepsilon-|u|)/\mu$. Since

$$\frac{(1-\varepsilon-|u|)(1+\varepsilon)}{1+\varepsilon-|u|} \le 1-\varepsilon \iff -|u|(1+\varepsilon) \le -|u|(1-\varepsilon),$$

that region lies inside $(1-\varepsilon)\mathcal{R}_{j-1}$, so the slice contributes at most $\mu^{j+1} M_{j,\varepsilon} I_j(H) = \mu M_{j,\varepsilon} I_j(F_u)$. For $j = 1$ use Cauchy–Schwarz, which gives $M_{1,\varepsilon} = 1+\varepsilon$. Integrate over u and use $\mu \le 1$.

Monotonicity in j follows by embedding: take $F(s,u) = H(s/\mu_\delta)\mathbf{1}_{[0,\delta]^{k-j}}(u)$ with $\mu_\delta = 1-(k-j)\delta/(1+\varepsilon)$ and let $\delta \to 0$. □

Slicing naively gives each slice its own parameter $\varepsilon' = \varepsilon/(1-|u|) \geq \varepsilon$, which would only bound the weighted constant by $\sup_{\varepsilon' \geq \varepsilon} M_{j,\varepsilon'}$, a weaker statement. The single rescaling by μ above avoids that loss.

Remark 11 (Computed, $k=3$). For the shifts $(0,1,2)$ and $D=1$ (so $\omega = 2$ and $\alpha_D(c) = 1$ for both weight vectors), with unrestricted polynomials (degree 5 for $k=3$, degree 10 for $k=2$, cut off at $|t| \leq \rho$ for $\rho \in \{1-\varepsilon, 1, 1+\varepsilon\}$; values truncated):

ε	$M_{2,\varepsilon} \geq$	$M_{3,\varepsilon}[(1,1,0)] \geq$	$M_{3,\varepsilon}[(\frac{1}{2},1,\frac{1}{2})] \geq$	$M_{3,\varepsilon} \geq$
0	1.385933	1.349844	1.160121	1.646427
1/10	1.518566	1.482142	1.264697	1.793492
1/5	1.627161	1.585674	1.341926	1.882281
1/3	1.717103	1.663743	1.413309	1.917001

Both weighted columns stay below $M_{2,\varepsilon}$, as the proposition requires. The data are in the file `notes/enlarged_pigeonhole_d5_d10.txt`.

Remark 12 (scope). Neither proposition covers more general polytope supports [15, §3], Stadlmann's hybrid supports [19], J-regions that depend on m, or arguments that are not linear in the indicators of primality. Those are Open.

4. Results of our own: how far the ε-enlargement alone can go

For the plain simplex the bound $M_k \leq \frac{k}{k-1}\log k$ shows where the method stops ($k \geq 51$). This section starts the same analysis for the ε-enlarged simplex. For $0 < \tau \leq L$ put

$$M_k(L,\tau) = \sup_F \frac{\sum_{m=1}^k \int_{\tau\mathcal{R}_{k-1}} \left(\int_0^\infty F\,dt_m\right)^2 \prod_{i\neq m} dt_i}{\int F^2},$$

the supremum over square-integrable $F \neq 0$ supported on $L\mathcal{R}_k$. Then $M_{k,\varepsilon} = M_k(1+\varepsilon, 1-\varepsilon)$ and $M_k = M_k(1,1)$. The quantity $M_k(L,\tau)$ is non-decreasing in L and in τ, satisfies $M_k(cL,c\tau) = c\,M_k(L,\tau)$, and is non-decreasing in k by the embedding used in the proof of Proposition 10. Write $S = t_1 + \cdots + t_k$ and call the index m *active* at t if $S - t_m \leq \tau$.

Lemma 13 (Proved here; it is [15, Lemma 6.1] with the region of the marginals kept)**.** *Let $\psi_1, \ldots, \psi_k$ be measurable functions on $L\mathcal{R}_k$, positive almost everywhere, such that for almost every $t' \in \tau\mathcal{R}_{k-1}$ and every m*

$$\int_0^{L-|t'|} \frac{du}{\psi_m(t',u)} \leq 1,$$

where (t',u) is the point whose m-th coordinate is u and whose other coordinates are t'. Then

$$M_k(L,\tau) \leq \operatorname*{ess\,sup}_{t\in L\mathcal{R}_k} \sum_{m \text{ active at } t} \psi_m(t).$$

Proof. For $t' \in \tau\mathcal{R}_{k-1}$ the Cauchy–Schwarz inequality gives

$$\left(\int F\,dt_m\right)^2 \leq \int \frac{dt_m}{\psi_m} \cdot \int \psi_m F^2\,dt_m \leq \int \psi_m F^2\,dt_m.$$

Integrating over $t' \in \tau\mathcal{R}_{k-1}$ gives the integral of $\psi_m F^2$ over the set where m is active. Sum over m. □

Proposition 14 (Proved here; elementary, possibly folklore)**.** *For $k \geq 2$ and $0 \leq \varepsilon < 1$,*

$$M_{k,\varepsilon} \leq (1+\varepsilon) M_k \leq (1+\varepsilon) \frac{k}{k-1} \log k.$$

Conversely, for $0 \leq \varepsilon \leq M_k/(4k)$,

$$M_{k,\varepsilon} \geq (1+\varepsilon) M_k - \frac{64k^2}{M_k} \varepsilon^2.$$

In particular $\lim_{\varepsilon \to 0^+} (M_{k,\varepsilon} - M_k)/\varepsilon = M_k$.

Proof. By monotonicity in τ and scaling, $M_k(1+\varepsilon, 1-\varepsilon) \leq M_k(1+\varepsilon, 1+\varepsilon) = (1+\varepsilon) M_k$, and $M_k \leq \frac{k}{k-1} \log k$ is [15, Cor. 6.4].

For the lower bound, scaling gives $M_{k,\varepsilon} = (1+\varepsilon) M_k(1, \rho)$ with $\rho = (1-\varepsilon)/(1+\varepsilon)$. Put $\delta = k(1-\rho)$ and let F on $\mathcal{R}_k$ satisfy $\sum_m J_k^{(m)}(F) \geq (M_k - \eta) I_k(F)$. Split $F = F_1 + F_2$ with $F_2 = F \mathbf{1}_{S > \rho}$, and let $x = I_k(F_2)/I_k(F)$. Two facts follow from the Cauchy–Schwarz inequality on segments of length at most $1-\rho$. First, a fiber of F_2 has length at most $1-\rho$, so $\sum_m J_k^{(m)}(F_2) \leq \delta\, I_k(F_2)$. Second, a fiber over t' with $|t'| > \rho$ lies in $\{S > \rho\}$ and has length at most $1-\rho$, so restricting the marginals of F to $\rho \mathcal{R}_{k-1}$ loses at most $\delta\, I_k(F_2)$. By the triangle inequality for $F \mapsto (\int F\, dt_m)_{m \leq k}$ and the first fact,

$$M_k - \eta \leq \left(\sqrt{M_k(1-x)} + \sqrt{\delta x} \right)^2 \leq M_k(1-x) + 2\sqrt{M_k \delta x} + \delta x.$$

This is a quadratic inequality in $\sqrt{x}$. If $\delta \leq M_k/2$ it gives $x \leq (16\delta + 8\eta)/M_k$. By the second fact the quotient of F for $M_k(1, \rho)$ is at least $M_k - \eta - \delta x$. Letting $\eta \to 0$ we get $M_k(1, \rho) \geq M_k - 16\delta^2/M_k$. Finally $\delta = 2k\varepsilon/(1+\varepsilon)$, and $\delta \leq M_k/2$ when $\varepsilon \leq M_k/(4k)$. □

So for fixed k the factor $1+\varepsilon$ is exact to first order, and no dual argument can improve it there. Any saving is an effect of large k: it needs ε of size about $1/k$ or more, where the shell $\rho < \sigma \leq 1$ carries most of the mass. The first inequality already restricts the parameter: with Bombieri–Vinogradov the criterion $M_{k,\varepsilon} > 4$ forces $\varepsilon > \varepsilon_0(k) = \frac{4(k-1)}{k \log k} - 1$ (Computed):

k	49	48	45	40	39	35	30
$\varepsilon_0(k)$	0.0068	0.0117	0.0274	0.0572	0.0638	0.0929	0.1368

For larger ε we use the lemma with a weight that changes shape where the indices start to become inactive.

Theorem 15 (Proved here for the inequality; Computed, interval arithmetic, for the numerical statement)**.** *Let $k \geq 2$, $0 < \tau < s_1 \leq L$ and $a \geq s_1$. Then*

$$M_k(L, \tau) \leq \sup_{0 \leq \sigma \leq \tau} W(\sigma), \qquad W(\sigma) = \frac{ka}{k-1} \log \left(1 + \frac{(k-1)(s_1 - \sigma)}{a - \sigma} \right) + L - s_1 + \sigma \log \frac{L - \sigma}{s_1 - \sigma}.$$

Consequently $M_{k,\varepsilon} < 3.99$ for every $k \leq 29$ and every $0 \leq \varepsilon < 1$. In particular the criterion of [15, Thm. 3.12] *on the ε-enlarged simplex cannot give* $\mathrm{DHL}[k, 2]$ *for any $k \leq 29$ from the Bombieri–Vinogradov theorem, nor from any level of distribution $\theta \leq 0.501$, whatever the value of ε.*

Proof. Fix $\Lambda > 0$, put $\lambda = \Lambda/(ka)$ and take the two-zone weight

$$\psi_m(t) = \lambda\, (a - S + k t_m) \quad (S \leq s_1), \qquad \psi_m(t) = \Lambda\, \frac{t_m}{S} \quad (S > s_1).$$

It is positive wherever $t_m > 0$. Where $S \leq s_1 \leq a$ every term is non-negative, so the sum over the active indices is at most the sum over all indices, which is $\lambda(k(a - S) + kS) = \Lambda$. Where

$S > s_1$ the sum over the active indices is at most $\Lambda \sum_m t_m/S = \Lambda$. On the fiber over t' with $|t'| = \sigma \le \tau$ we have $S = \sigma + u$, and

$$\int_0^{s_1-\sigma} \frac{du}{\lambda(a-\sigma+(k-1)u)} + \int_{s_1-\sigma}^{L-\sigma} \frac{(\sigma+u)\,du}{\Lambda u} = \frac{W(\sigma)}{\Lambda}.$$

Lemma 13 with $\Lambda = \sup W$ gives the inequality.

For the numerical statement it suffices to treat $k = 29$. For $\varepsilon \le 143/1000$, Proposition 14 gives $M_{29,\varepsilon} \le 1.143 \cdot \frac{29}{28} \log 29 < 3.9863$. The interval $[143/1000, 1]$ is cut into 857 intervals $[e, e']$ of length $1/1000$. For each one the file `code/cert_floor_k29.json` lists rational a and s_1 with $\sup W < 3.99$ for $(L, \tau) = (1+e', 1-e)$; by monotonicity in L and τ this covers every $\varepsilon \in [e, e']$. The supremum over σ is bounded rigorously: the first logarithm is non-increasing in σ (because $a \ge s_1$) and the last term is non-decreasing, so on a piece $[x, y]$ we bound W by the first term at x plus the last at y, in ball arithmetic at rational points, bisecting until the bound is below 3.99. The script is `code/dual_eps_cert.py`; a second script, `code/check_floor_cert.py`, re-checks the certificate with a different interval library. The largest value of $\sup W$ over the 857 intervals is 3.9772 (floating point). □

Remark 16 (what was known, and a caution). Polymath8b state the uniform bound $M_{k,\varepsilon} \le \frac{k}{k-1} \log(2k-1)$ for $0 \le \varepsilon < 1$ [15, Prop. 6.5], and remark that the bounds for $M_{k,\varepsilon}$ "do not appear to be as sharp". That bound is below 4 exactly for $k \le 23$, so Theorem 15 moves this floor from 24 to 30. It does not use their bound. We note, with caution, that we could not complete the last step of the proof of [15, Prop. 6.5] as we read it: the weights $1 - S + kt_m$ are summed, and for $S > 1$ their sum over the active indices can exceed k (at $t = (s, 0, \dots, 0)$ with $1 < s \le 1 + \varepsilon$ only the first index is active and its weight is $1 + (k-1)s$). Our reconstruction gives the constant $\frac{k}{k-1} + \varepsilon$ in place of $\frac{k}{k-1}$. For version 0.4 we re-read the proof in the current arXiv text: after the Cauchy–Schwarz step it says only "Integrating in $t_1, \dots, t_{i-1}, t_{i+1}, \dots, t_k$ and summing in i, we obtain the claim." The weights $1 - S + kt_i$ of the inactive indices can be negative when $S > 1$, and dropping them is what lets the sum exceed k; it is at most $k + (k-1)(S-1) \le k + (k-1)\varepsilon$, which gives our constant. So the argument as printed proves $M_{k,\varepsilon} \le \left(\frac{k}{k-1} + \varepsilon\right) \log(2k-1)$. Whether the stated inequality is nevertheless true is OPEN; we have not seen the journal version, and we may be missing an intended step. A brief search (the paper [15] and the Polymath wiki page on the variational problem) found no sharper published upper bound for $M_{k,\varepsilon}$; the search was not exhaustive, and Theorem 15 has not been reviewed by anyone outside this project.

Remark 17 (COMPUTED (FLOATING POINT): how far this weight goes). The best value of $\sup W$ over (a, s_1), maximised over ε, is about 3.975 for $k = 29$, 4.004 for $k = 30$, 4.257 for $k = 40$ and 4.459 for $k = 50$, in each case near $\varepsilon = 0.24$ to 0.28; the bound of [15, Prop. 6.5] is 4.187, 4.218, 4.482 and 4.689. For small ε the two-zone weight does not beat the scaling bound: at $k = 50$ and $\varepsilon = 1/25$ both give 4.15 (4.1458 against 4.1515), while Polymath8b's certified lower bound is 4.0043. The true threshold is presumably close to $k = 50$, so the theorem is far from sharp. Script: `code/dual_eps.py remark15`.

Remark 18 (another weight; whether it beats scaling is OPEN). We also tried the weight $\psi_m = \frac{\beta}{n}(a - S_A + nt_m)$, where n is the number of active indices at t and S_A the sum of the active coordinates. It is positive for every $a > \tau$ (an active index has $t_m \ge S - \tau$, so $a - S_A + nt_m \ge a - \tau$). Its sum over the active indices is exactly βa everywhere, so the bound is a times the largest fiber integral. Evaluating the fiber integral exactly on the fibers over $t' = (x, \dots, x, 0, \dots, 0)$ only gives a lower bound for what this weight achieves. Version 0.12 did this for $a \in [1.05, 3]$ only (`code/dual_eps.py remark16`, which still scans that grid), found 4.178 at the edge $a = 1.05$,

and concluded wrongly that the weight is worse than scaling. Rerunning the same fiber code over $a > \tau$ (COMPUTED (FLOATING POINT), $k = 50$, $\varepsilon = 1/25$) gives about 4.14633 at $a = 1.037$, with binding fiber $r = 10$ non-zero coordinates, below the 4.1515 from scaling (4.16437 at $a = 1.03$, 4.15020 at $a = 1.04$, 4.17821 at $a = 1.05$). This reproduces the value 4.146 of an earlier draft (version 0.3), which came from fibers with about ten non-zero coordinates. Since the test only bounds what the weight achieves from below, whether this weight, with all fibers taken into account, beats the scaling bound is OPEN.

The limit of weights that see only t_m and S (version 0.4). Every weight used so far has the form $\psi_m(t) = \Psi(S - t_m, t_m)$: Polymath8b's $\lambda(a - S + kt_m)$ and both zones of Theorem 15. Call this family $\mathcal{W}$. For weights in $\mathcal{W}$ the problem "minimise the bound of Lemma 13" is convex: each fiber condition $\int du/\Psi \le 1$ is convex in Ψ, and each point t gives a linear condition $\sum_{m \text{ active}} \Psi(S - t_m, t_m) \le \Lambda$. So its optimum can be approximated, and the approximation suggests how far a weight in $\mathcal{W}$ can go.

Proposition 19 (COMPUTED (FLOATING POINT), discretised; not a rigorous bound in either direction; HEURISTIC)**.** *Discretise $\Psi = \rho \cdot (L - \sigma + (k-1)u)$ with ρ constant on cells of an $N \times$ (geometric) grid in (S, u), impose the fiber conditions exactly on the grid $\sigma \in \frac{L}{N}\mathbb{Z}$, and impose the pointwise condition on all configurations of the grid of step L/N (found by a knapsack oracle). The convex optimum $B_N(k, \varepsilon)$, computed by cutting planes with a floating-point conic solver (Clarabel) at each step, is:*

(k, ε)	$N = 80$	$N = 120$	$N = 160$	*best weight known*	$M_{k,\varepsilon} \ge$
$(50, 0)$	*3.9919*			*3.9919 (Cor. 6.4)*	*3.9274*
$(50, 1/25)$	*4.1427*	*4.1420*	*4.1418*	*4.1458 (Thm. 15)*	*4.0043*
$(30, 1/4)$	*4.0501*	*4.0345*	*4.0240*	*4.0021 (Thm. 15)*	–

At $\varepsilon = 0$ the optimum over the whole family is the classical weight: the solver, started anywhere, returns $\frac{k}{k-1}\log k$. At $(50, 1/25)$ the discretised optimum settles near 4.14. This suggests (HEURISTIC) that no weight that sees only t_m and S can bring the dual bound much below 4.14 there, while the true value is at least 4.0043; but B_N is neither a restriction nor a relaxation of the problem over $\mathcal{W}$, and at $(30, 1/4)$ it exceeds the value 4.0021 of a known member of $\mathcal{W}$ by about 0.022. At $(30, 1/4)$ the discretisation has not converged (the values still fall, and lie above the continuum value ≤ 4.0021 that Theorem 15 already certifies), so we draw no conclusion about the exact floor of $\mathcal{W}$; runs at $k = 31, 32$ gave $B_{120} = 4.0261$ to 4.0976, also unconverged. The lesson, heuristically, is the first one: *to approach the truth near $k = 50$, a dual weight must depend on more of t than t_m and S.* The optimal weight is known in principle: if $F > 0$ is the extremal function, $\psi_m = \int F\, dt_m / F$ makes Lemma 13 an equality (the Collatz–Wielandt principle for the positive operator $F \mapsto \sum_{m \text{ active}} \int F\, dt_m$). A first test (COMPUTED (FLOATING POINT), 20000 random points, not a bound; `code/cw_sample.py`) is encouraging on the plain simplex and discouraging on the enlarged one. For $k = 3$, $\varepsilon = 0$ and the degree-12 Galerkin function ($M_3 \ge 1.646440$) the sampled ratio stays in $[1.6464, 1.6465]$, which would be an almost sharp upper bound if the supremum could be certified. For $\varepsilon > 0$ the Galerkin polynomial changes sign ($\min F / \max F = -0.0003$ at $(k, \varepsilon, d) = (10, 1/5, 12)$, -0.11 at $(50, 1/25, 12)$). This is structural: where no index is active the extremal function vanishes, so near that region any positive F makes the ratio large. Making the principle usable there is OPEN. The script is `code/dual_lp.py`; `runs/dual_lp_conv.log` and `runs/dual_lp_floor.log` hold the output, including the solver's warning "Solution may be inaccurate" on some of the runs in the table; the $N = 80$ values at $(50, 1/25)$ and $(50, 0)$ are not in these logs. The last column is from Table 2 and Theorem 6.

Remark 20 (heuristic: does the enlarged simplex stop at $k = 50$?). The certified values of Table 2 grow with the degree roughly geometrically. Fitting $M(D) = M_\infty - Cr^D$ to three consecutive even degrees (Aitken's Δ^2, script `code/extrapolate_eps.py`) can be tested where the answer is known. From the COMPUTED values at $D = 16, 18, 20$ for $k = 50$, $\varepsilon = 1/25$ it predicts 3.99324, 4.00195, 4.00553 at $D = 22, 25, 27$, against the certified 3.99310 (degree 22), our 4.00124 (degree 25) and Polymath8b's 4.0043 (degree 27). The model is slightly optimistic, by 0.0001 to 0.0012. Applied to $k = 49$ with $D = 18, 20, 22$ (certified 3.97714 and 3.97689 at degree 22 for $\varepsilon = 1/30$ and $1/25$) it gives limits 3.9922 and 3.9938. As a test registered before the run, the fit predicted 3.98230 at degree 24 for $\varepsilon = 1/30$; the certified value is 3.98218 (1508 coefficients), again slightly below, and the fit through $D = 20, 22, 24$ moves the limit down to 3.9913. The grid $\varepsilon \in \{1/50, 1/40, 1/30, 1/25, 1/20\}$ at $D \le 20$ gives at most 3.995. This suggests that $\sup_\varepsilon M_{49,\varepsilon} < 4$, so that the ε-enlarged simplex with Bombieri–Vinogradov stops at $k = 50$, consistent with the fact that every bound below 246 uses a different support. This is a heuristic, not a theorem: OPEN. Together with Theorem 15 it brackets the floor of this method between 30 and (conjecturally) 50.

Remark 21 (scope). Theorem 15 concerns the ε-enlarged simplex only. It says nothing about the 2026 claims, which use other supports and new equidistribution estimates. A floor for those is the Problem in the next section.

5. STADLMANN-TYPE SUPPORTS (VERSIONS 0.5–0.9)

This section takes the first steps on the next rung, Stadlmann's region for $H_1 \le 240$ [19], and gives a first rigorous answer to the Problem in the Programme (Section 6) for the one-band supports used by [19] and [3]. For this version a helper re-read both sources (the arXiv text of [19], the draft [3]); our notes with page references are in `notes/stadlmann-recheck-s5.md`.

The support and the criterion. The following is LITERATURE (PREPRINT), read in [19, Def. 1, Def. 5, Prop. 1]; the implication from the criterion to DHL$[k, 2]$ is CLAIMED. For $\delta > 0$, $-\varepsilon = A_0 < A_1 < \cdots < A_n < \frac{1}{2} - \varepsilon$ and caps $\delta < B_{j,m} \le B_{j,m+1} \le B_{j,m} + \delta$, the support $T_k(\delta, \underline{A}, \underline{B}, \varepsilon)$ is the set of $t \in [0,1]^k$ such that, for the j with $\sum_i t_i \in [A_{j-1} + \varepsilon, A_j + \varepsilon)$, one has $\sum_{i \in I} t_i \le B_{j,|I|}$ where $I = \{i : t_i > \delta\}$ (the "large" coordinates). With one band ($n = 1$), the functional $J(F)$ of [19] is the integral of $\left(\int F\, dt_k\right)^2$ over $t_1 + \cdots + t_{k-1} \le A_1 - \varepsilon$ (our reduction of the general definition), $I(F) = \int_{T_k} F^2$, and the criterion is $k(1 - c_1)J(F) - kc_2K(F) > I(F)$ for a symmetric F supported on T_k, where $c_1, c_2 \ge 0$ come from the minorant and $K(F) \ge 0$. Stadlmann's choice is $k = 49$, $\varepsilon = 0.0075$, $A_1 = 0.253$, $\delta = 0.028$, $B_{1,1} = B_{1,2} = 0.15$, $B_{1,m} = 0.17$ ($m \ge 3$) and $c_1 = c_2 = 0$; that of [3] is $k = 45$, $\varepsilon = 1/125$, $A_1 = 257/1000$, $\delta = 41/2500$ with caps listed in their Table 3. Both use one band. The equidistribution estimates behind the criterion are CLAIMED; we have not checked them.

In the Polymath scaling $u = 4t$ the support lies in $L\mathcal{R}_k$ with $L = 4(A_1 + \varepsilon)$, the marginals are integrated over $\tau\mathcal{R}_{k-1}$ with $\tau = 4(A_1 - \varepsilon)$, and the criterion with $c_1 = c_2 = 0$ reads $kJ/I > 4$. Write $s = (L + \tau)/2 = 4A_1$ and $e = (L - \tau)/(L + \tau) = \varepsilon/A_1$; the scaling $M_k(cL, c\tau) = cM_k(L, \tau)$ of Section 4 gives

$$M_k(L, \tau) = s\, M_k(1 + e, 1 - e) = 4A_1\, M_{k,\varepsilon/A_1}. \tag{2}$$

For Stadlmann, $(L, \tau) = (1.042, 0.982)$, $s = 253/250$, $e = 15/506$; for [3], $(L, \tau) = (1.06, 0.996)$, $s = 257/250$, $e = 8/257$.

Without the caps.

Proposition 22 (Computed, exact rational arithmetic)**.** *With even power sums,* $M_{49,15/506} \geq 3.9685534$ *(degree* 20*,* 707 *coefficients) and* $M_{45,8/257} \geq 3.9029887$ *(degree* 21*,* 846 *coefficients). Hence, by* (2)*,* $M_{49}(1.042, 0.982) \geq 4.016176$ *and* $M_{45}(1.06, 0.996) \geq 4.012272$*.*

The two certificates, in the directory `code/`, are `cert_stad49_capfree_D20e.json` and `cert_ax45_capfree_D21e.json`; the script `verify_cert.py` re-checks them. Our degree-21 basis spans the space $\mathcal{B}_{21}$ of [3, Thm. 11.1] (Literature (preprint)): their §11 defines $\mathcal{B}_{21}$ as the span of $(L - P_1)^a P_\lambda$ with P_λ a product of even power sums and $a + |\lambda| \leq 21$, with the same $L = 53/50$ (definition quoted in `notes/sources-enlarged-supports.md`), which is our family; its dimension, 846, agrees with theirs. On their capped support they report the exact value $4.0043840983\ldots$ (Claimed); on the larger cap-free region, with the same space, we get 4.012272. Both are lower bounds, for M_T and for $M_{45}(1.06, 0.996) \geq M_T$ respectively, so the comparison proves nothing; but it suggests (if their value is right) that at this degree the caps cost about 0.008. This is a comparison, not a verification of [3].

With the caps. For $F = P\,1_T$ with P a polynomial, write $I(P)$, $J(P)$ for the cap-free functionals on $L\mathcal{R}_k$ (marginals over $\tau\mathcal{R}_{k-1}$) and $V = L\mathcal{R}_k \setminus T$ (Polymath scaling; $g = 4\delta$, $C_m = 4B_{1,m}$).

Proposition 23 (Proved here; elementary)**.** *Let* $R_m(g, c) = \{u \in L\mathcal{R}_k : \ u_1, \ldots, u_m > g,\ u_1 + \cdots + u_m \geq c\}$*, put* $c_m = C_m$ *for* $m \leq 6$ *and* $c_7 = 7g$*. Then, with*

$$E(P) = \sum_{m=1}^{7} \binom{k}{m} \int_{R_m(g,c_m)} P^2,$$

one has $I_T(F) \leq I(P)$ *and, for every* $0 < \eta < 1$*,*

$$J_T(F) \ \geq \ \max\left(J(P) - 2\sqrt{J(P)\,L\,E(P)},\ (1-\eta)J(P) - (\eta^{-1} - 1)L\,E(P)\right).$$

Proof. A point of V has a large set I with $\sum_{i\in I} u_i > C_{|I|}$. If $|I| \leq 6$ it lies in a copy of $R_{|I|}(g, c_{|I|})$; if $|I| \geq 7$, any seven of its large coordinates exceed g and sum to more than $7g$, so it lies in a copy of $R_7(g, c_7)$. By symmetry of P^2, $\int_V P^2 \leq E(P)$. On the fiber over u' write $a = \int P\,du_k$ and $b = \int P\,1_V\,du_k$, so that the marginal of F is $a - b$. Cauchy–Schwarz gives $b^2 \leq L \int P^2 1_V\,du_k$, hence $\int b^2 \leq L\,E(P)$. Then $(a-b)^2 \geq a^2 - 2|a||b|$ with Cauchy–Schwarz once more, or $(a-b)^2 \geq (1-\eta)a^2 - (\eta^{-1} - 1)b^2$. Finally $I_T(F) = \int_T P^2 \leq I(P)$. □

The integrals over $R_m(g, c)$ are again exact rational numbers for the basis $(L - P_1)^a P^b$ (Proved here; `code/caps_exact.py`). Write the first m coordinates as $g + z$. Integrating out the other $k - m$ coordinates with Lemma 3 leaves powers of $L - mg - |z|$; the shifted power sums are $P_j(g + z) = \sum_l \binom{j}{l} g^{j-l} P_l(z)$ with $P_0 = m$ and $P_1 = |z|$; at fixed $r = |z|$ the integral of $\prod_l P_l(z)^{f_l}$ is $r^{m-1+w(f)} E_m(f)/(m - 1 + w(f))!$ (now with odd parts as well); and the remaining integral in r over $[\max(c - mg, 0), L - mg]$ is elementary. The code reproduces Lemma 3 exactly when $g = c = 0$, and agrees with direct symbolic integration (sympy) for $k = 3$, $m = 1, 2$, $g = 1/10$.

Since all three quantities are exact quadratic forms in the coefficients of P, we choose P as the top eigenvector of $(1 - \eta)J - (\eta^{-1} - 1)L\,G_E$ against I, where G_E is the Gram matrix of E, and evaluate both bounds exactly. The penalty works: at degree 16 it lowers E/I from $3.2 \cdot 10^{-4}$ (for the cap-free optimiser) to $3.4 \cdot 10^{-7}$.

Proposition 24 (Computed, exact rational arithmetic)**.** *For Stadlmann's parameters ($k = 49$) there is a polynomial P of degree 20 in even power sums (707 coefficients) with*

$$\frac{kJ_T(P\,1_T)}{I_T(P\,1_T)} \;\geq\; 3.95972.$$

At degree 16 (307 coefficients) the same method gives 3.93625.

This is below 4, and in version 0.5 the rung 240 was therefore not reproduced; Theorem 26 below supersedes it. The certificate is `code/cert_stad49_caps_D20e.json`, written by `code/stadlmann_caps_opt.py`; the best bound came from $\eta = 1/800$, with $E/I = 1.6 \cdot 10^{-7}$ and the bound of Proposition 23 in its square-root form. Where the loss comes from (Computed, same runs): the cap-free optimiser of degree 20 has $s\,kJ/I = 4.01618$ but $E/I = 1.2 \cdot 10^{-3}$ (of which $1.2 \cdot 10^{-4}$ from points with one coordinate above 0.6); forcing E down costs about 0.045 of the cap-free ratio ($4.0161 \to 3.9710$), and the Cauchy–Schwarz step costs another 0.011. Both losses come from the way we bound J_T, not from the caps as such: $F = P\,1_T$ is cut off on V anyway, and Proposition 23 charges for P on V only because it does not compute the capped marginals. The next step is therefore an exact formula for $J_T(P\,1_T)$, which needs the fibers of T and not only integrals over V. A first attempt that dropped the condition $u_i > g$ in the union bound failed badly ($E/I \approx 146$ for the cap-free optimiser; `runs/stad49_caps_D20.log`).

The capped functionals exactly (version 0.6). Version 0.5 bounded $J_T(P\,1_T)$ through Proposition 23, and the Cauchy–Schwarz step there turned out to be the whole difficulty. We now compute $I_T(P\,1_T)$ and $J_T(P\,1_T)$ exactly. Notation as above: u in the Polymath scaling, $R = L - |u'|$ on the fiber over $u' \in \mathbb{R}^{k-1}_{\geq 0}$, $I(u) = \{i : u_i > g\}$, $m(u) = |I(u)|$, $s(u) = \sum_{i \in I(u)} u_i$, and $Q(u', x) = \int_0^x P(u', t)\,dt$.

Proposition 25 (Proved here; elementary)**.** *Assume $C_{m+1} \leq C_m + g$ for all m (this is Stadlmann's condition $B_{j,m+1} \leq B_{j,m} + \delta$) and $g \geq L - \tau$ (this holds for every parameter set below: $L - \tau = 0.06$ against $g = 0.112$ and 0.0728 for [19, 8], and $L - \tau = 0.064$ against $g = 0.0656$ for [3]). Call u'* admissible *if $s(u') \leq C_{m(u')}$. Then:*

1. *the fiber $\{t \geq 0 : (u', t) \in T\}$ is empty if u' is not admissible, and otherwise equals $[0, U(u')]$ with $U = \min\big(R, \max(g,\, C_{m(u')+1} - s(u'))\big)$;*
2. *one has $I_T(P\,1_T) = I(P) - \int_V P^2$ and*

$$J_T(P\,1_T) = J(P) - \mathrm{Corr}_1 - \mathrm{Corr}_2,$$

   where $\mathrm{Corr}_1 = \int Q(u', R)^2$ over the non-admissible u' with $|u'| \leq \tau$, and $\mathrm{Corr}_2 = \int \big(Q(u', R)^2 - Q(u', U)^2\big)$ over the admissible u' with $U < R$;
3. *each of the three correction integrals is a finite signed sum, over the size m of the large set and over $j \geq 0$ further coordinates forced above g (inclusion–exclusion for "all other coordinates are at most g", with weight $(-1)^j \binom{n}{m}\binom{n-m}{j}$), of integrals over polygons in the plane of $r_A = s - mg$ and $\rho = |u| - s - jg$. On each polygon the integrand, after the shape integrals of Lemma 3 and of Proposition 23, is a polynomial in (r_A, ρ), so every term is an exact rational number.*

For Stadlmann's parameters every correction piece has $m \leq 9$ and $j \leq 3$.

Proof. (1) If $t \leq g$ the large set of (u', t) is $I(u')$, so $(u', t) \in T$ iff u' is admissible and $|u'| + t \leq L$. If $t > g$ the large set is $I(u') \cup \{k\}$ and the condition is $s(u') + t \leq C_{m(u')+1}$; since $C_{m+1} \leq C_m + g$, this condition with $t > g$ implies $s(u') < C_{m(u')}$, so a non-admissible u' has an empty fiber. Hence for admissible u' the fiber is $[0, \min(R, g)] \cup (g, \min(R, C_{m+1} - s)]$, which is $[0, U]$. (2) $F = P\,1_T$ has marginal $Q(u', U)$ on admissible u' and 0 otherwise, while the cap-free marginal is $Q(u', R)$;

subtract. (3) By symmetry, integrate over the points whose large set is exactly $A = \{1, \dots, m\}$ and multiply by $\binom{n}{m}$; the condition that the remaining coordinates are at most g is $\prod(1-1_{u_i>g})$, expanded. On such a piece put $z = u - g$ on the $m+j$ forced coordinates; at fixed $r_A = |z_A|$ and $\rho = |z_B| + |y|$ (y the free coordinates) the integral of a product of power sums is a sum over splits of the product of the A-group density of Proposition 23 and the convolution of the j shifted coordinates with the free ones, which is again of Dirichlet type. The integrand (P^2, $Q(R)^2$, $Q(g)^2$ or $Q(x)^2$ with $x = C_{m+1} - s$) is a polynomial in R and x, both affine in (r_A, ρ). The regions are: $r_A \geq C_m - mg$ and $r_A + \rho \leq$ const (the non-admissible pieces); $r_A \geq C_{m+1} - (m+1)g$ and $R > g$ (where $U = g$); and the rectangle $r_A < C_{m+1} - (m+1)g$, $\rho < L - C_{m+1} - jg$ (where $U = C_{m+1} - s$). In the last two cases $U < R$ forces $R > g$, so $|u'| < L - g \leq \tau$ by the hypothesis $g \geq L - \tau$, and these fibers lie in the domain of J. In every piece the large set carries at least $C_1 = 0.6$ or the free mass is below $L - C_1$, so $jg < 0.442$, $j \leq 3$; and $mg < L$ gives $m \leq 9$. These one-line inequalities are PROVED HERE; their integer-unit analogues, for lists of natural numbers in units of 1/1000, are LEAN-CHECKED (`stadlmann_extra_le_three`, `stadlmann_large_count`). □

Theorem 26 (COMPUTED, exact rational arithmetic, two independent programs)**.** *Let P be the cap-free optimiser of degree* 20 *of Proposition 22 (*707 *coefficients, even power sums). For Stadlmann's parameters ($k = 49$, $L = 1.042$, $\tau = 0.982$, $g = 0.112$, $C_1 = C_2 = 0.6$, $C_m = 0.68$ for $m \geq 3$) and $F = P\,1_T$,*

$$\frac{k\,J_T(F)}{I_T(F)} = 4.0120389035\ldots \;>\; 4.$$

The polynomial, in u-coordinates, is `code/cap6/P_capfree_D20_u.json` (written from the certificate `code/cert_stad49_capfree_D20e.json` by `code/to_ucoords.py`; the cap-free value kJ/I it gives equals $\frac{253}{250}$ times the certified Rayleigh quotient exactly). Two programs compute I_T and J_T as exact rationals and agree on both numbers digit for digit: `code/capped_exact.py` (ours; Proposition 25, "cap-free minus corrections", 46 minutes on three cores) and `code/capped_check.py`, written by a helper from the definitions only, without access to our code, which sums directly over the exact large set (no cap-free term at all) with exponential-moment generating functions (6 minutes on four cores). The checks behind them: the direct and the corrected I_T in `capped_exact.py` agree exactly; a Monte Carlo test at $k = 12$, $D = 4$, where the caps remove 47% of I and 45% of J, agrees to within one standard deviation (`code/cap6/mc_test.py`); and `capped_check.py` agrees exactly with a brute-force evaluation by polytope triangulation and symbolic integration for $k = 2, \dots, 5$ with caps that bite (`code/capped_check_tests.py`). Results: `code/cap6/res_capfree_D20.json`, `runs/capped_check_D20.json`.

The caps cost only 0.0041: relative to the cap-free values, $\int_V P^2 = 1.209 \cdot 10^{-3}\, I(P)$, and Corr_1, Corr_2 on the pieces $U = g$ and $U = C_{m+1} - s$ are $1.72 \cdot 10^{-4}$, $8.1 \cdot 10^{-5}$ and $1.98 \cdot 10^{-3}$ of $J(P)$. The largest loss is the cut of the fiber at $C_{m+1} - s$ (for $m = 0$: no coordinate above 0.6). So Proposition 24 (3.9597) measured the weakness of Proposition 23, not the caps; it is superseded. The loss grows with the degree of the cap-free optimiser (COMPUTED, exact: `capped_check.py` for every degree; the degree-20 value also with `capped_exact.py`, and the degree-8 value too, though that run's output is not stored in the repository):

degree	8	12	16	20
cap-free kJ/I	3.789069	3.923943	3.987180	4.016176
capped kJ_T/I_T	3.789068	3.923849	3.985976	4.012039

so higher-degree optimisers put more mass near the caps; optimising P on T itself (done in version 0.8, Proposition 32) can only raise the capped value. At $k = 45$ this is consistent with the comparison after Proposition 22, where the caps of [3] appeared to cost about 0.008.

Corollary 27 (conditional; the analytic input is CLAIMED)**.** *The variational step of* [19] *is independently reproduced: the criterion of* [19, Prop. 1] *holds for $k = 49$ with his support, his caps and his parameters, and $c_1 = c_2 = 0$. Together with the* LEAN-CHECKED *admissible* 49*-tuple of diameter* 240 *(Theorem 1), $H_1 \le 240$ then follows from his reduction* [19, Prop. 1] *and his equidistribution theorem, neither of which we have verified.*

This is a reproduction of part of a claimed result, not a new bound, and it says nothing about the correctness of the new equidistribution estimates of [19] (his Lemmas 3–7 and Prop. 3), on which everything rests. Our formulation agrees with [19] as re-read in our notes (`notes/stadlmann-recheck-s5.md`), with these differences, none of which changes the value: we use the scaling $u = 4t$ (criterion $kJ/I > 4$); we use our own reduction of his J for one band (the band conditions on t_k and t_k' follow from the support of F); we ignore his closed versus half-open band endpoints (measure zero); and we use our own test space (even power sums times powers of $L - P_1$, degree 20) instead of his $\mathcal{B}_{19}$ or $\mathcal{B}_{21}$, whose coefficients he does not print.

Remark 28 (COMPUTED, interval arithmetic: a ceiling, our own). Every coordinate of a point of T is at most 0.6 (PROVED HERE, by the same linear inequality; an integer-unit analogue in units of $1/1000$ is LEAN-CHECKED: `stadlmann_coord_le`) and $T \subset L\mathcal{R}_k$, so the capped quantity is at most $M_{49}(1.042, 0.982)$, which Theorem 15 with $a = 1.0417$, $s_1 = 1.0007$ bounds by 4.14 (certified with `dual_eps_cert.certify_box`, 337 pieces). Hence $4.0120 < \sup_F kJ_T/I_T \le 4.14$. Cutting the fibers of the two-zone weight at 0.6 lowers its floating-point ceiling only from 4.13785 to 4.13753: the supremum of W sits near $\sigma = \tau$, where the fibers are short anyway, so a ceiling that sees the caps needs a weight that sees the large coordinates (OPEN).

The next rung: the support of [3] (version 0.7). The same computation applies to the one-band support of [3] ($k = 45$, $H(45) = 212$), whose caps change with m. In the Polymath scaling their Table 3 gives $L = 53/50$, $\tau = 249/250$, $g = 41/625$ and $C_m = 4B_{1,m}$ with $B_{1,m} = \frac{1}{5000}(777, 794, 875, 917, 953, 983, 1016, 1042, 1063, 1081)$ for $m \le 10$ and $B_{1,m} = 1081/5000$ for $m \ge 11$ (LITERATURE (PREPRINT), read in [3]; our notes record the page). These caps satisfy $\delta < B_{1,m} \le B_{1,m+1} \le B_{1,m} + \delta$ (LEAN-CHECKED: `charton_caps_valid`), and $g = 0.0656 \ge L - \tau = 0.064$. At most 16 coordinates can exceed g, and every correction piece of Proposition 25 has $j \le 6$ (PROVED HERE; the integer-unit analogues in units of $1/1250$ are LEAN-CHECKED: `charton_large_count`, `charton_extra_le_six`).

Proposition 29 (COMPUTED, exact rational arithmetic, two independent programs)**.** *Let P be our cap-free optimiser of degree* 21 *for $M_{45}(1.06, 0.996)$ (Proposition 22,* 846 *coefficients) and T the support of* [3] *with the caps of their Table 3. Then*

$$\frac{45\, J_T(P\,1_T)}{I_T(P\,1_T)} = 4.0027351128\ldots \;>\; 4.$$

The value was computed by `code/capped_check.py` (13 minutes on four cores; input `code/cap6/P_ax45_D21_u.json`, result `code/cap6/check_ax45_D21.json`). Our own program `code/capped_exact.py` (Proposition 25, now with caps depending on m; 108 minutes on four cores, 228 pieces, result `code/cap6/res_ax45_D21.json`) gives the same I_T and J_T as exact rationals. Relative to the cap-free values, $\int_V P^2 = 2.67 \cdot 10^{-3} I(P)$ and the three J corrections are $3.0 \cdot 10^{-4}$, $9.6 \cdot 10^{-5}$ and $4.65 \cdot 10^{-3}$ of $J(P)$. The cap-free value of the same P is 4.012272, so the caps cost 0.0095 here, more than at $k = 49$: g is smaller (0.0656 against 0.112), so more coordinates count as large, and the largest correction, the cut at $C_{m+1} - s$, is $4.65 \cdot 10^{-3}$ of

$J(P)$ against $1.98 \cdot 10^{-3}$. The value reported in [3, Thm. 11.1], with their own optimised F in the same polynomial space, is 4.00438 (CLAIMED); ours is lower, as it should be for a vector optimised without the caps, and it already exceeds 4.

As recorded in version 0.5, the draft [3] is not consistent about its caps: its Lemma 9.1 states $C(1) = C(2) = 31/50$ and $C(s) = 17/25$ for $s \geq 3$. With those caps the same P gives

$$\frac{45\, J_T(P\,1_T)}{I_T(P\,1_T)} = 3.9736869862\ldots \;<\; 4$$

(COMPUTED, exact, `capped_check.py`, `code/cap6/check_ax45_D21_lemma91caps.json`). This does not show that the criterion fails for those caps: our P was optimised without caps, and the caps of Lemma 9.1 are strictly smaller than those of Table 3 for every $m \geq 1$, so their support is smaller. But it does show that, with this test function, the conclusion depends on which of the two cap sets the draft intends, and that Table 3 is the one under which the criterion is easy to confirm. Within our 846-dimensional space a test function optimised on the smaller support does not reach 3.99 (Proposition 33, version 0.9); whether a test function outside that space can work with Lemma 9.1's caps is OPEN.

We do *not* lower the project's bound to 212 in this version. The analytic input of [3] (its equidistribution statements at level $\omega = 7/1000$ for moduli generated by these caps) is CLAIMED, the draft is not consistent about which caps it uses, and with the other cap set no test function in our space reaches 3.99 (Proposition 33). What is established here is narrower: *for the caps of Table 3 of* [3]*, the variational criterion of* [3, Thm. 4.10] *holds at* $k = 45$ (COMPUTED, two programs). Together with the LEAN-CHECKED 45-tuple of diameter 212, this would give $H_1 \leq 212$ conditionally on the analytic input of [3] for those caps; whether that input covers the caps of Table 3 is the question an expert reader of [3] should settle first.

Remark 30 (COMPUTED, interval arithmetic: ceiling for $k = 45$). As in Remark 28, Theorem 15 with $a = 1.0596$, $s_1 = 1.0166$ gives $M_{45}(1.06, 0.996) \leq 4.126$ (495 pieces). So the capped value of [3] lies in (4.0027, 4.126]. Cutting the fibers at the largest coordinate $C_1 = 0.6216$ moves the floating-point ceiling of this weight only from 4.12481 to 4.12446.

Optimising the test function on the capped support (version 0.8). Up to version 0.7 every capped value came from a polynomial optimised *without* the caps and then truncated to T. Both capped functionals are quadratic forms in the coefficients c of P: $I_T(P\,1_T) = c^\top G_I c$ and $J_T(P\,1_T) = c^\top G_J c$.

Proposition 31 (COMPUTED; a description of the method, validated by the exact self-tests below, not a proved statement)**.** *In the direct method of* `code/capped_check.py`*,* I_T *is a linear function of the coefficient products* $\kappa_I[F](y) = \sum_{F_1+F_2=F} p_{F_1}(y)p_{F_2}(y)$ *and* J_T *a linear function of* $\kappa_J[F](y,v) = \sum_{F_1+F_2=F} q_{F_1}q_{F_2}$*, where* $P = \sum_F p_F(y) \prod_j P_j^{F_j}$*,* $y = L - P_1$*, and* $q_F(y,v)$ *are the coefficients of* $\int_0^v P\,dt$*. Running the chain of exact linear maps of that program backwards gives exact weights* $w_I[F][\alpha] = \partial I_T/\partial \kappa_I[F][y^\alpha]$ *and* $w_J[F][(b,\gamma)] = \partial J_T/\partial \kappa_J[F][y^b v^\gamma]$*, and then* $G_I[t_1,t_2] = w_I[F_1+F_2][a_1+a_2]$ *for basis terms* $t_i = (L-P_1)^{a_i}\prod_j P_j^{F_{i,j}}$*, and* $G_J = Q^\top H Q$ *with* $H[(F_1,b_1,\gamma_1),(F_2,b_2,\gamma_2)] = w_J[F_1+F_2][(b_1+b_2,\gamma_1+\gamma_2)]$ *and* Q *the (sparse, explicit) map from basis terms to the* q_F*.*

The reverse pass costs about one forward evaluation (`code/capped_gram.py`). Its self-test is exact: for the input polynomial, $\sum\langle \kappa_I, w_I\rangle$ and $\sum\langle \kappa_J, w_J\rangle$ equal the I_T and J_T of `capped_check.py` as rationals (passed at $k = 12$, at $k = 49$ with the polynomial of Theorem 26, and at $k = 45$ with the polynomial and caps of Proposition 33). G_I is exact; G_J is assembled in fixed point with an explicit entrywise error bound. The matrices are only used to *find* a

vector, or, with that error bound, to certify that $\sigma G_I - kG_J$ is positive definite (recursive Schur complements in ball arithmetic), i.e. that no P in the whole test space exceeds σ; every capped value stated as a certificate is recomputed from the rounded vector by the exact evaluators.

Proposition 32 (COMPUTED, exact rational arithmetic, two programs)**.** *For Stadlmann's support and parameters ($k = 49$, as in Theorem 26) let P^* be the top generalised eigenvector of $(k\,G_J, G_I)$ in the same 707-dimensional space (even power sums, degree 20), rounded to 400 significant bits per coefficient. Then*

$$\frac{k\,J_T(P^*1_T)}{I_T(P^*1_T)} = 4.0124198475\ldots \;>\; 4.$$

Files: `code/cap6/P_opt49_D20r_u.json` (vector), `code/cap6/check_opt49_D20r.json` (`capped_check.py`, 7 minutes on two cores), and `code/cap6/res_opt49_D20r.json` (`capped_exact.py`, 60 minutes on two cores); the two programs agree on I_T and J_T as exact rationals. The unrounded vector gives the same value to twelve digits (`check_opt49_D20.json`). The cap-free value of P^* is 4.0158303, against 4.0161761 for the cap-free optimiser.

So optimising on T regains only 0.00038 of the 0.0041 that truncation cost (about 9%): in this test space the margin of the variational step of [19] is 0.0124, and the caps cost 0.0038 even for the best polynomial of degree 20.

Proposition 33 (COMPUTED, ball arithmetic, certified at $\sigma = 3.99$; it rests on the Gram weights of one program (`capped_check.py` run backwards), checked at one random point (Schwartz–Zippel test; the second exact program's run at that point is not finished); versions 0.8–0.9)**.** *Let T' be the support of [3] at $k = 45$ with the caps of their Lemma 9.1 ($C_1 = C_2 = 31/50$, $C_m = 17/25$ for $m \geq 3$; $L = 53/50$, $\tau = 249/250$, $g = 41/625$), and let V be our 846-dimensional space (even power sums times powers of $L - P_1$, degree ≤ 21). Then $3.99\,G_I - 45\,G_J$ is positive definite, so*

$$\frac{45\,J_{T'}(P\,1_{T'})}{I_{T'}(P\,1_{T'})} < 3.99 < 4 \qquad \text{for every } 0 \neq P \in V.$$

The weights w_I, w_J of Proposition 31 were computed exactly (88 minutes on two cores, version 0.8). G_J was assembled with absolute error at most 2^{-P} per weight, and the resulting entrywise error bound was put into the balls. Positive definiteness was certified by recursive Schur complements in ball arithmetic: for $\sigma = 4$ at $P = 5000$ bits with halving blocks (version 0.8), and again for $\sigma = 4$ and $\sigma = 3.99$ at $P = 6500$ bits with blocks split $1 : 2$ (version 0.9; `code/capped_gram.py --ceiling 4/1,3990/1000 --split 1/3`, log `runs/gram45_lemma91_recheck.log`).

How the weights were checked. Both $c \mapsto c^\top G_I c$ and the map $c \mapsto I_{T'}$ computed by `capped_check.py` are polynomials of degree 2 in the 846 coefficients, and likewise for J. If they differed, their difference would be a nonzero polynomial of degree at most 2. By the Schwartz–Zippel lemma it then vanishes at a uniformly random point of S^{846}, $|S| = 2^{32}+1$, with probability at most $2/|S| < 2^{-31}$. At the random integer vector `code/cap6/P_rand45_lemma91_u.json` (seed recorded in the file) the weights reproduce $I_{T'}$ and $J_{T'}$ of `capped_check.py` *exactly* (`runs/gram45_rand_selftest.log`, `code/cap6/check_rand45_lemma91.json`). The ball forms $c^\top G_I c$ and $c^\top G_J c$ contain these exact values. So the proposition rests on `capped_check.py` (which agrees exactly with the independent `capped_exact.py` at Stadlmann's and at Charton et al.'s Table 3 parameters, Theorem 26 and Proposition 29) up to a failure probability below 2^{-31} for each functional. The same random point has not yet been evaluated by `capped_exact.py`. That run needs about four hours here and was stopped after 153 of its pieces (`runs/exact_rand45_lemma91.log`), so the cross-check of `capped_check.py` itself at these caps is still to come. The exact weights used for both tests were loaded from a file outside the repository and are not yet archived.

The optimum. Inverse iteration in V converges to $45J_{T'}/I_{T'} = 3.9855478$ (`code/cap6/P_opt45_lemma91_D21r_u.json`, the eigenvector rounded to 400 bits; COMPUTED (FLOATING POINT), a value from the Gram matrices, not re-certified exactly). So in V the truncated cap-free vector (3.97369) is beaten by 0.0119, the best possible lies (up to that floating-point evaluation) in [3.98554, 3.99), and 4 is out of reach.

Since our space V is the space $\mathcal{B}_{21}$ of [3, Thm. 11.1] (see after Proposition 22), this suggests that the caps of Lemma 9.1 cannot give their 4.00438, and that the value they report belongs to the caps of their Table 3, for which our truncated vector already gives 4.0027 (Proposition 29). The ceiling rests on one program and a randomised check, and we have not compared their support and fiber conventions with ours in every detail, so this is a located discrepancy for the authors or an expert reader, not a refutation.

Remark 34 (a failed prediction, our own). Before the run we registered a prediction (`notes/session8-prediction.md`). At $k = 49$, optimising on T had regained only 9% of what truncation cost (0.00038 of 0.0041). We guessed that re-optimisation is a second-order effect and would again regain about a tenth, giving 3.977–3.981 here. It regains 31% (0.0119 of 0.0386), and the optimum 3.9855478 lies outside the predicted range. A pure second-order picture, with gain about loss^2/gap and the gap fitted at $k = 49$ (gap $\approx 0.0041^2/0.00038 \approx 0.044$), would instead predict a gain of about 0.033, nearly the whole loss. Neither guess is right: the fraction regained grows with the size of the truncation loss, but not quadratically. The heuristic is ruled out as a quantitative tool.

The rung 236: the support of Kintali (versions 0.8–0.9). [8] uses Stadlmann's criterion with one band at $k = 48$ and the support "D182" (LITERATURE (PREPRINT), read in [8, §2]): $\varepsilon = 3/400$, $A = 253/1000$, $\delta = 91/5000$ and a single cap $B_m = 1899/10000$ for every $m \geq 1$. In our scaling this is $L = 1.042$ and $\tau = 0.982$ (Stadlmann's values), $g = 91/1250$ and $C_m = 1899/2500$ for all m; it satisfies $C_{m+1} \leq C_m + g$ and $g \geq L - \tau$, so our evaluators apply unchanged (`code/to_ucoords.py --params kintali`). At most ten coordinates of a point of this support exceed g, and every correction piece has $j \leq 3$ (PROVED HERE: in the cap-violation pieces and in the Corr_2 pieces alike, $jg < L - C_1 = 0.2824$, while $4g = 0.2912$; integer-unit analogues in units of $1/2500$ are LEAN-CHECKED: `kintali_large_count`, and `kintali_extra_le_three` for the cap-violation pieces only). The analytic input is Stadlmann's (his Prop. 3, with the Type IIc range made empty); the test function is a public 1295-term witness of degree 25 (the $k = 50$ certificate of the AxiomMath `PrimeGapsLib` repository, according to [8, §3]), transported by the dilation $z = \frac{26}{25U}t$ and used at $k = 48$ without re-optimising; the claimed values are $48J/I = 1.0007747\ldots$ in his normalisation, i.e. 4.0030988 in ours, and 4.0036348 without the caps (CLAIMED). The second 236 preprint [18], whose full text was supplied to us after version 0.8, uses the same support with the same witness (LITERATURE (PREPRINT), read in [18, §4]; notes in `notes/sources-236-song-yue.md`). Its criterion divides by the cap-free denominator, $48J(F)/I_0 > 2001/2000$. The authors state that a fresh regeneration of their capped numerator is still pending. An exact recomputation of it from their table is a natural next step for us.

Proposition 35 (COMPUTED, exact rational arithmetic, two independent programs; versions 0.9 and 0.12)**.** *Let P be our cap-free optimiser of degree 22 for $M_{48,15/506}$ (1041 coefficients, even power sums; certified Rayleigh quotient 3.9589046383, so $kJ/I = \frac{253}{250} \cdot 3.9589046 = 4.0064115$ without caps), rounded to 400 significant bits per coefficient, and T the support of [8] above. Then*

$$\frac{48\, J_T(P\,1_T)}{I_T(P\,1_T)} = 4.0060814208\ldots \;>\; 4.$$

Computed by `code/capped_check.py` (22 minutes on four cores; input `code/cap6/P_kin48_D22r_u.json`, result `code/cap6/check_kin48_D22r.json`; certificate `code/cert_kin48_capfree_D22e.json`). In version 0.12 our second program, `capped_exact.py` (cap-free value minus 176 inclusion–exclusion pieces, checkpointed; 15.5 core-hours in two segments: it was resumed at 77 of 176 pieces after a worker restart, and the second segment took 2.1 hours on four cores), gave the same I_T and J_T as exact rationals (`code/cap6/res_kin48_D22r.json`, `code/compare_two_programs.py`, `code/cap6/compare_kin48_D22r.json`). At degree 20 the same construction stays below 4 (3.9994 even without caps).

Kintali's caps are generous: they cost our vector only 0.00033, against 0.0041 for Stadlmann's caps at $k = 49$. Our value is above his claimed 4.0031, which is plausible because his witness was optimised for $k = 50$ and a different support and then transported, while ours is optimised for this k and this (L, τ). So the variational step of [8] (Stadlmann's criterion for this support at $k = 48$) holds with our own test function. This is *not* a verification of 236: the analytic input (Stadlmann's equidistribution estimates, with the corrections described in [8]) is Claimed and unverified by us.

The witness of Song and Yue, recomputed exactly (version 0.10). Appendix A of [18] prints the witness as 1295 integer triples (a, α, c), $P(z) = \sum c\,(\frac{26}{25} - s(z))^a m_\alpha(z)$ with monomial symmetric functions m_α and $a + |\alpha| \le 25$, used as $P_U(t) = P(\frac{26}{25U}t)$ on the support above ($U = \frac{521}{2000}$, $k = 48$). Appendix C prints reference integers $I_{\mathrm{ref}} = (98!)^2 I_0^*$ and $J_{\mathrm{ref}} = (99!)^2(26!)^2 J^*$, where I_0 is the *cap-free* mass on $\{s < U\}$, J the capped numerator, and the stars denote the integer coordinates $x = 10^4 t$. The criterion is $48J/I_0 > \frac{2001}{2000}$ (Literature (preprint), read in [18, §4.12, App. C]; that it implies DHL$[48, 2]$ is Claimed). The authors write that a fresh evaluation reproduces I_{ref}, while a fresh regeneration of J_{ref} from the table "remains incomplete".

All parts of every α in the table are even. In our coordinates $u = 4t$ we have $z = \frac{520}{521}u$ and $\frac{26}{25} - s(z) = \frac{520}{521}(L - P_1(u))$, and the augmented monomials $A_\beta = |\mathrm{Aut}\,\beta|\, m_\beta$ satisfy $p_r A_\beta = A_{(\beta,r)} + \sum_i A_{\beta + re_i}$. So P_U is, exactly, a polynomial in $L - P_1$ and the even power sums $P_2, \dots, P_{24}$ (1311 terms, degree 25; `code/songyue_convert.py`, checked exactly at random rational points of $\mathbf{R}^{48}$ by `code/songyue_pointcheck.py`).

Proposition 36 (Computed, exact rational arithmetic; the capped numerator by two independent programs)**.** *For the witness P_U of* [18] *and the support T above:*

(1) $(98!)^2\, 10^{192}\, 5^{150}\, 521^{50}\, I_0 = I_{\mathrm{ref}}$ *exactly;*
(2) $(99!)^2(26!)^2\, 10^{196}\, 5^{150}\, 521^{50}\, J_T = J_{\mathrm{ref}}$ *exactly, where J_T is the capped numerator;*
(3) *hence* $48J_T/I_0 = 1.000664921591\ldots > \frac{2001}{2000}$, *i.e.* $4.0026596864\ldots$ *in our scaling. With the capped denominator,* $48J_T/I_T = 4.0030988047\ldots$ *($I_T/I_0 = 0.99989031$). Without the caps,* $48J_0/I_0 = 4.0036348167\ldots$.

The factor $10^{192}5^{150}521^{50}$ (resp. $10^{196}5^{150}521^{50}$) is what the change to the integer coordinates $x = 10^4 t$ gives when $Q = 65125^{25} P_U$ (our reading of [18, Eq. (71)]; note $65125 = 25 \cdot 2605$ and $z = 26x/65125$), so items (1) and (2) are equalities of integers with 655 and 715 digits, not numerical agreements. Item (1) comes from a program written for this purpose, `code/songyue_I0.py`. It works in the monomial basis of the table: each pair $m_\alpha m_\beta$ is integrated by summing over partial matchings of the parts, with a bitmask dynamic programme. It does not use the power-sum conversion. J_T in (2) is `code/capped_check.py` on the converted polynomial (56 minutes on four cores; `code/cap6/check_sy236.json`; comparison `code/songyue_compare.py`, output `code/cap6/compare_sy236.json`). J_0 is again in the monomial basis (`code/songyue_J0.py`). The values 4.0030988 and 4.0036348 agree with the capped and cap-free values claimed in [8] to every printed digit.

Second program (version 0.11). `code/capped_exact.py` (Proposition 25: cap-free value minus the correction integrals Corr_1, Corr_2, here 126 inclusion–exclusion pieces with $m \le 13$, $j \le 3$, each an exact rational, checkpointed piece by piece) was run on the same converted polynomial with the option `--jonly` (numerator only; about 32 core-hours, resumed from a checkpoint of five pieces). Its J_T equals the J_T of `capped_check.py` as an exact rational, hence also $(99!)^2(26!)^2\,10^{196}\,5^{150}\,521^{50}\,J_T = J_{\mathrm{ref}}$ (`code/songyue_compare2.py`, outputs `code/cap6/compare2_sy236.json`, `code/cap6/compare_sy236_exact.json`); its cap-free I and J equal the monomial-basis values I_0, J_0 of `songyue_I0.py` and `songyue_J0.py` exactly. The two programs for J_T share only the input file: one sums over the exact large set, the other subtracts corrections from the cap-free value. The corrections are small: $\mathrm{Corr}_1/J = 2.3 \cdot 10^{-6}$, $\mathrm{Corr}_2/J = 2.4 \cdot 10^{-4}$, and the pieces with $j = 3$ further coordinates above g are below $10^{-23}J$ in absolute value although they take almost half of the running time (15.3 of 32.2 core-hours, 47%; `runs/exact_sy236_jonly.log`, `code/cap6/pieces_sy236_jonly.jsonl`).

So the capped numerator of [18], whose fresh regeneration the authors list as pending, is reproduced exactly by two independent implementations, and the variational criterion of both 236 preprints holds for their published witness. This is a statement about the variational step only. The equidistribution input ([18, Theorem 4.18]; Stadlmann's estimates as amended in [8]) and the reductions from it to the variational criterion ([18, Thm. 4.33, §§4.12–4.14]; [19, Prop. 1]) are CLAIMED and not verified by us.

Corollary 37 (conditional; the analytic input is CLAIMED)**.** *The variational step of* [18] *is independently reproduced: the criterion* $48J/I_0 > \frac{2001}{2000}$ *of* [18, Prop. 4.21] *holds for their witness, support and parameters at* $k = 48$*, by two independent exact programs (Proposition 36). Together with the* LEAN-CHECKED *admissible* 48*-tuple of diameter* 236 *(`boundedGaps_236_of_DHL_48`),* $H_1 \le 236$ *then follows from their reduction* [18, Theorem 4.33 and §§4.12–4.14] *and their equidistribution theorem* [18, Theorem 4.18]*, neither of which we have verified.*

This is a reproduction of part of a claimed result, not a new bound. The project's reproduced bound becomes 236 with exactly this caveat. The witness is theirs; our own degree-22 vector on the same support (Proposition 35, 4.0061) is also confirmed by two programs since version 0.12.

Remark 38 (an audit of the analytic input of 236; CLAIMED, audited inside this project, not peer review). In version 0.13 we record the outcome of an adversarial check of the proof of [18, Theorem 4.18] (§§4.1–4.7, and its use in §§4.13–4.14), carried out inside this project by AI auditors, one per part of the proof, with exact-rational checks of every exponent inequality and a word-for-word comparison of the cited statements of Polymath8a [14] (`notes/audit-sy236-thm418/`, scripts in `code/audit_sy236/`). Its verdict: the theorem appears to follow from the cited literature after one local repair. The displayed bound (49) of [18, p. 27] for the correlation of two Kl_3 sums, transcribed correctly from [14, Lemma 6.25, Cor. 6.26], carries the factor $\gcd(g, m_1 - m_2)^{1/2}$ with $g = (r_1, r_2)$, $r_i = gt_i$; the prime-by-prime proof gives $\gcd(g, m_1t_1^3 - m_2t_2^3)^{1/2}$ instead (the local correlation is of size p exactly when the two local arguments coincide, where it equals $\sum_a |\mathrm{Kl}_3(a;p)|^2$). After this change the sum over m_1, m_2 has the same bound, and no exponent or margin changes. We re-ran the auditors' numerical check (`code/audit_sy236/kl3_coordinator_check.py`, $p = 31$, $t_1 = 2$, $t_2 = 3$): the normalised correlation is at most 1.03 when $m_1 = m_2$ and at least 29.97 when $m_1t_1^3 \equiv m_2t_2^3 \pmod p$. The auditors also found that the swapped-factor step of Remark 41 does not occur in the proof of [18]. This does *not* change the status of 236: the audit is one AI reading per section, not expert review; the deep inputs (Deligne's bounds, bilinear Bombieri–Vinogradov, Siegel–Walfisz) stay LITERATURE; and §4.12, the Selberg main term and Lemma 4.32 of [18] were not audited, so the reduction [18, Thm. 4.33, §§4.12–4.14] as a whole remains unverified. The bound 236 remains conditional (Corollary 37).

One-sided certificates without the thin pieces (version 0.12). Exact equality of two programs is the strongest check, but the criterion itself only needs a lower bound for J_T and an upper bound for I_T. The following observation lets a one-sided certificate omit the most expensive pieces of Proposition 25(3).

Proposition 39 (PROVED HERE; an instance of the Bonferroni inequalities, so folklore in substance)**.** *In Proposition 25(3) write* $\mathrm{Corr}_1 = \sum_{j\ge 0}(-1)^j S_j$ *and* $\int_V P^2 = \sum_{j\ge 0}(-1)^j S'_j$*, where* S_j *(resp.* S'_j*) is the sum over* m *of* $\binom{n}{m}\binom{n-m}{j}$ *times the piece with* j *forced coordinates. Then* $S_j, S'_j \ge 0$*, and for every* $t \ge 0$

$$\sum_{j\le 2t+1}(-1)^j S_j \;\le\; \mathrm{Corr}_1 \;\le\; \sum_{j\le 2t}(-1)^j S_j,$$

and the same for S'_j*. In particular* $J_T \ge J - (S_0 - S_1 + S_2) - \mathrm{Corr}_2$ *and* $I_T \le I - S'_0 + S'_1$*. A certificate of* $kJ_T/I_T > 4$ *therefore needs only the pieces of* Corr_1 *with* $j \le 2$*, those of* $\int_V P^2$ *with* $j \le 1$*, and all pieces of* Corr_2*.*

Proof. Fix m and $A = \{1, \dots, m\}$ and let $G_A(u) \ge 0$ be the integrand of the piece ($Q(u', R)^2$ times the indicator of $|u'| \le \tau$ and $s_A > C_m$ for Corr_1; P^2 times the indicator of $|u| \le L$ and $s_A > C_m$ for $\int_V P^2$). On the set where every coordinate of A exceeds g, the indicator that no other coordinate exceeds g is $\prod_{i\notin A}(1 - x_i)$ with $x_i = 1_{u_i > g} \in \{0, 1\}$. If N of the x_i equal 1, then $\sum_{j\le\ell}(-1)^j e_j(x) = \sum_{j\le\ell}(-1)^j\binom{N}{j} = (-1)^\ell\binom{N-1}{\ell}$ for $N \ge 1$ (and 1 for $N = 0$), which is $\ge \prod(1 - x_i)$ for even ℓ and $\le$ it for odd ℓ. Multiply by $G_A \ge 0$, integrate, and sum over m and over the $\binom{n}{m}$ choices of A; by symmetry the integral of $G_A\, e_j(x)$ is $\binom{n-m}{j}$ times the piece with j forced coordinates, which is ≥ 0. The last two inequalities are the cases $t = 1$ and $t = 0$ together with $I_T = I - \int_V P^2$ and $J_T = J - \mathrm{Corr}_1 - \mathrm{Corr}_2$. □

Size of what is saved (COMPUTED, from the exact pieces of Proposition 36, `code/cap6/pieces_sy236_jonly.jsonl`). On the witness of [18] the four levels of Corr_1 are $2.3 \cdot 10^{-6}J$, $-1.7 \cdot 10^{-10}J$, $7.0 \cdot 10^{-20}J$ and $-2.5 \cdot 10^{-69}J$ for $j = 0, 1, 2, 3$, and the $j = 3$ level took 7.1 of the 32.2 core-hours; all 46 exact pieces of Corr_1 there, and the 96 pieces of Corr_1 and $\int_V P^2$ of the second program for our own $k = 48$ vector (`code/cap6/pieces_kin48_D22r.jsonl`), have the sign $(-1)^j$, as the proposition requires. For that vector the one-sided certificate of the proposition gives $48J_{\mathrm{low}}/I_{\mathrm{up}} = 4.006081420807 > 4$, $3.5 \cdot 10^{-12}$ below the exact ratio, and the omitted pieces took 4.3 of its 15.5 core-hours (28%). The loss can also be bounded a priori (`code/thin_bound.py`): on $\{|u| \le L\}$ one has $0 \le R \le L$ and $0 \le P_\ell(u) \le |u|^\ell$, so $|P| \le M := \sum_i |c_i| L^{a_i + w(b_i)}$ and $|Q| \le LM$; and the region of a $j = 3$ piece of Corr_1 has $C_m - mg \le r_A \le r_A + \rho \le \tau - (m + 3)g$, an interval of length $\tau - C - 3g = \frac{1}{250}$ for the caps of [8, 18], whatever m is. The resulting bound is $1.4 \cdot 10^{-14}J$, and it is above all ten exact $j = 3$ values. The same crude bound is useless for Corr_2 (it gives $6 \cdot 10^{32}J$): there the corresponding length is $L - C - 3g = \frac{8}{125}$, sixteen times larger in about 44 dimensions, and the coefficient bound M is far above the actual size of $|P|$ (cancellation in the power-sum basis). Both bounds above are for the witness of [18] (`code/cap6/thin_bound_sy236.json`); for a different polynomial, our own $k = 48$ vector, we sampled $|P|$ and found M larger by a factor of about 10^{28} (that sampling is not stored in the repository). Since the integrand of Corr_2 has no sign, the proposition does not apply to it directly; splitting it into its $Q(u', R)^2$ and $Q(u', U)^2$ parts and applying the proposition to each is the next thing to try (done in version 0.13, Proposition 40).

Proposition 40 (Proved here; Bonferroni again, folklore in substance)**.** *Write* $\mathrm{Corr}_2 = \mathrm{Corr}_2^R - \mathrm{Corr}_2^U$*, where, with* $W = \{u' \text{ admissible}, U < R\}$*,*

$$\mathrm{Corr}_2^R = \int_W Q(u',R)^2, \qquad \mathrm{Corr}_2^U = \int_W Q(u',U)^2,$$

and expand each, as in Proposition 25(3), as $\sum_{j\ge 0}(-1)^j S_j^R$ *and* $\sum_{j\ge 0}(-1)^j S_j^U$ *over the number* j *of forced coordinates. Then* $S_j^R, S_j^U \ge 0$ *and*

$$J_T \;\ge\; J - (S_0 - S_1 + S_2) - \left(S_0^R - S_1^R + S_2^R\right) + \left(S_0^U - S_1^U\right), \qquad I_T \le I - S_0' + S_1'.$$

So a one-sided certificate of $kJ_T/I_T > 4$ *needs no piece with* $j \ge 3$*, and no piece of* Corr_2^U *or of* $\int_V P^2$ *with* $j \ge 2$*.*

Proof. In both cases of Proposition 25(3) (whether $U = g$ or $U = C_{m+1} - s_A$), the region of a piece and the value of U are functions of u' that depend on the set A of large coordinates only through $m = |A|$ and s_A (and on u' through R), not on the indicators $1_{u_i > g}$ of the other coordinates; given A, the integrands $Q(u',R)^2$ and $Q(u',U)^2$ are ≥ 0. So the proof of Proposition 39 applies verbatim to each of the two parts: the truncation after an even level is an upper bound and after an odd level a lower bound. Use the upper bound at level 2 for Corr_1 and Corr_2^R and the lower bound at level 1 for Corr_2^U in $J_T = J - \mathrm{Corr}_1 - \mathrm{Corr}_2^R + \mathrm{Corr}_2^U$. □

The one-sided certificate on the 240 *vector* (Computed, exact; `capped_exact.py --bonf`; files `code/cap6/res_capfree_D20_bonf.json`, `pieces_capfree_D20_bonf.jsonl` and `compare_capfree_D20_bonf.json` there). For the cap-free optimiser of Theorem 26 on the support of [19] ($k = 49$), the 98 pieces that the proposition keeps give

$$49\, J_{\mathrm{low}}/I_{\mathrm{up}} = 4.012038868083\ldots > 4,$$

$3.5 \cdot 10^{-8}$ below the exact ratio 4.0120389035705 of Theorem 26; $J_{\mathrm{low}} \le J_T$ and $I_{\mathrm{up}} \ge I_T$ hold as exact rationals, and every kept level has the sign $(-1)^j$. The two parts of Corr_2 are each about 35 times larger than their difference (case $U = C_{m+1} - s$: $7.03 \cdot 10^{-2} J$ and $6.83 \cdot 10^{-2} J$ at $j = 0$), but their levels still fall fast enough: the largest omitted level is of order $10^{-8} J$, far below the margin $3 \cdot 10^{-3}$. With the split, no piece with three forced coordinates is needed at all. This is a cheaper route to the same conclusion, not a new one.

Remark 41 (on [19, Lemma 5]; Literature (preprint) reading, not a result). Song and Yue [18, Remark 4.23] say that one step in the proof of [19, Lemma 5] lacks a hypothesis. We read the step in arXiv v1. Lemma 5 (Baker–Irving Type I) takes $f = \alpha \star \beta$ with α an arbitrary coefficient sequence and β smooth at scale x^γ. For $\gamma \in (\frac{1}{2}, \frac{1}{2} + 2\omega + \varepsilon]$ the proof applies Lemma 3 (Polymath Type II) "with the roles of α and β swapped". Lemma 3 requires its factor at scale $\le x^{1/2}$ to have the Siegel–Walfisz property, and after the swap that factor is α. Neither Lemma 5 nor case (I) of [19, Definition 9], where Lemma 5 is used, assumes this property of α. So the step reads as [18] describe it. A natural repair is to check that the sequences α produced by the decomposition do have the property, but we have not checked this, and we do not know whether the conclusion is affected. Our results on 240 and 236 concern only the variational step and are unaffected either way. The *bound* 240 may depend on this step, and so may the routes to 236 of [8] (which uses [19, Prop. 3]) and to 212 of [3]; we have not traced either. Our route to 236 goes through [18, Theorem 4.18] (Corollary 37), and the internal audit of its proof found no swapped-factor step there (Remark 38).

Remark 42 (the next rung, 186; Literature (preprint) reading and one Lean-checked fact). We read [13] and its numerical companion (the file `short_gaps_numerics.pdf` of the repository `openai/PrimeGaps186`) in version 0.11. Their criterion is not a capped simplex ratio

of the kind treated above. The sieve has one outer sum and, for each coordinate, two nested inner sums, and the criterion [13, Theorem 4.5] is a "signed minorant" inequality. It involves $I = \|F\|^2$, the inner masses $J_{0,i}$ and $J_{+,i}$, and a residual $E_{0,i}$ weighted by the negative part of Stadlmann's prime minorant, at $k = 40$. The test function lives on "fragment configurations": each coordinate carries a Poisson process with intensity dv/v, so coordinate totals have Dickman densities. It is a 77-coefficient rational step function on a grid of 98304 cells per coordinate. Their Lemma 2.5 states $\rho^* J^-/I^+ > \frac{500103}{500000}$ and a loss bound $L^+/I^- \le 696075110 \cdot 10^{-12}$ (from 97 components). We checked only the final rational arithmetic of their (2.34)–(2.35) (exact; it holds as printed), not any integral. The 40-tuple of [13, (1.3)] is admissible of diameter 186 (LEAN-CHECKED: `boundedGaps_186_of_DHL_40'`; the project already had another such tuple). Reproducing the variational step of 186 therefore requires a new evaluator: the fragment-measure forms, and the loss cover for the support failures. The pipeline of this paper does not cover it. In version 0.11 we transcribed the forms I_H and $J_{\lambda,H}$ of the companion and its table of 77 coefficients (`notes/sources-186-forms.md`). Two facts shape any reproduction. First, the cell masses of the Dickman densities are transcendental, so an independent check can only be an interval computation, not an exact rational one. Second, by symmetry everything reduces to one-dimensional arrays of length about 10^5, so the cost is a few thousand long polynomial products at a few hundred bits (our estimate). One formula of the companion, its layered cap convolution, is not specified completely in the text. The analytic inputs (a triply densely divisible variant of Stadlmann's bilinear estimate, the prime minorant, and the exponential-sum bounds that are axioms of their Lean development) remain CLAIMED. In version 0.12 we made a first independent check of one ingredient (COMPUTED (FLOATING POINT), 40 digits, not interval arithmetic; `code/oai186_dickman_cells.py`). On $[0,3]$ the Dickman function has closed forms, $\rho_D(u) = 1 - \log u$ on $[1,2]$ and $\rho_D(u) = 1 - \log u + \frac{1}{2}\log^2 u + \mathrm{Li}_2(1/u) - \frac{1}{2}\log^2 2 - \mathrm{Li}_2(\frac{1}{2})$ on $[2,3]$, and so does $\int \rho_D$. With these we evaluated every cell mass of the seven grid caps of the companion (363,450 cells; `code/oai186_dickman_cells.json`). All lie inside the enclosures given by its recurrence and cell bound, as we transcribed them, and the enclosures have width at most $8.35 \cdot 10^{-10}$ in units of the cell. One transcription doubt (which of two bars is the upper one) turns out not to matter at this precision. This checks one input of the companion; it is not yet an evaluation of I_H or $J_{\lambda,H}$.

Proposition 43 (COMPUTED, two independent programs, the second in ball arithmetic; version 0.13)**.** *Evaluate the forms of* [13]*'s companion, as transcribed in* `notes/sources-186-forms.md`*, with the layered cap measure of (2.15) read as follows (our derivation): split the retained* 39 *coordinates by the layer* $(b_{a-1}, b_a]$ *of their largest fragment over all seven cap values* b *that occur in the outer and face domains, and admit the erased coordinate in the outer shells whose cap is* $\ge b_a$*. Then*

$$I_H = 2.36853178533\ldots \cdot 10^{-14}, \qquad J_{\lambda,H} = 9.0248755655\ldots \cdot 10^{-14},$$

$$\frac{\rho^* J_{\lambda,H}}{I_H} = 1.000206087\ldots > \frac{500103}{500000},$$

with $J_0 = 9.00605408 \cdot 10^{-14}$*,* $J_+ = 1.91277 \cdot 10^{-16}$ *and* $J_t = 1.61669 \cdot 10^{-15}$*. Our* I_H *lies inside the enclosure* $[23685317816, 23685317890] \cdot 10^{-24}$ *of* [13, Lemma 2.5]*, and our* $J_{\lambda,H}$ *is* $5.9 \cdot 10^{-9}$ *(relative) above its lower bound* $90248755123 \cdot 10^{-24}$*.*

How it was computed (`code/oai186_IH.py`, `code/oai186_J.py`, outputs `code/oai186_IH.json`, `code/oai186_J.json`; 7 and 23 minutes on one core). Nothing from the authors' repository was used or run. By symmetry each form is a sum over the radial index $r < n = 98264$ of exact rational weights against coefficients of products of one-dimensional arrays. $M_{d,\eta} = \sum_\pi (d)_{|\pi|} P_0^{d-|\pi|} \prod_{B\in\pi} P_{e(B)}$ is summed over the set partitions π of the labels of the signature η;

we checked this formula against brute force on small cases. The Dickman function was evaluated through its closed forms (logarithm and Li_2), and the profile and Z in 110-digit floating point. The coordinate arrays were rounded down and up at scale 2^{-256}. All products are exact integer products by Kronecker substitution in GMP, not FLINT, which is the library of [13]. The erased-coordinate correlations are exact, and the final signed sums are exact integer sums. The result is therefore accurate far beyond the 10^{-9} of the comparison, but it is not a rigorous interval enclosure (the special functions are not interval-evaluated). A fresh-context audit of the derivation and both scripts found no defect. A second program (`code/oai186_check2.py`, combined by `code/oai186_check2_combine.py`; output `code/oai186_check2.json`) works in ball arithmetic throughout (python-flint `arb`, with the Dickman closed forms evaluated in balls) and obtains the coordinate sums differently, as $m!\,[y^m]A(y)^d$ for the exponential generating series $A(y) = \sum_j K_j X^j \exp\left(\sum_e y_e t_j^e\right)$, instead of by set partitions. It gives $I_H \in [2.36853178533302 \cdot 10^{-14} \pm 3 \cdot 10^{-30}]$, $J_{\lambda,H} \in [9.02487556547131 \cdot 10^{-14} \pm 4 \cdot 10^{-29}]$ and $\rho^* J_{\lambda,H}/I_H \in [1.00020608684 \pm 4 \cdot 10^{-12}]$, agreeing with the first program to $7 \cdot 10^{-17}$ (relative). Both programs use the same transcription and the same reading of (2.15), so they check the arithmetic and the code, not the reading; the agreement with the published values checks the reading.

What this does and does not show. The main-term quotient of [13, Lemma 2.5] is reproduced independently, and so is the reading of its layered cap convolution that we had marked as underspecified. Not reproduced: the loss bound $L^+/I_H \le 696075110 \cdot 10^{-12}$ (the 97 support-failure components of their §2.4), which the criterion also needs; and, as always, the analytic inputs (Claimed). The project's bound does not move.

Source data and the bookkeeping of the loss bound (Computed, exact rational arithmetic; version 0.14; the full companion was supplied to us after version 0.13). (i) From the rational definitions in [13, (1.13)–(1.19), (1.23)–(1.24)] we recomputed the two source ladders (29 and 43 rows, with order counts $(12, 12, 5)$ and $(12, 12, 19)$), the retained rows (0–27 and 0–38) and the four boundary values $\lfloor B/h \rfloor$, the continuous caps and, through the grid rule, the integer cap masks (1.5) on which our I_H and $J_{\lambda,H}$ rest (we had taken them as given), and the six group parameters with every guard of their Lemma 1.1. Every printed value is reproduced (`code/oai186_support_data.py`). The analytic comparisons in that lemma, with the distribution inequalities of [13], are not checked. (ii) For the 97 component bounds of their Tables 2.1–2.6 (`code/oai186_tables_2_1_to_2_6.json`) every Young inequality $cA + B/c \le E$ and inner weighting, the six group totals, the total $696075110 < 697 \cdot 10^6$, the component counts against their witness schedule, and the final margins (2.34)–(2.35) hold exactly (`code/oai186_tables_check.py`). This checks how the raw bounds are combined, not the raw bounds themselves; those come from the forms (2.16)–(2.21) of the companion, which we have transcribed (`notes/sources-186-loss.md`) but not yet evaluated.

Remark 44 (the last rung, 182; Literature (preprint) reading only). In version 0.13 we read [16] (a 132-page bundle: the main paper and two analytic companions; notes in `notes/sources-182.md`). It claims DHL$[39, 2]$ and hence $H_1 \le 182$ (its Theorem 1.2). Its framework is the fragment-measure framework of [13] on a grid four times finer (393216 cells), with four cap domains, a profile of 726 rational coefficients (the factor g of [13] and its 11 power-sum signatures times 22 cubic B-splines), a new five-prime minorant and a support restoration step. The claimed surplus before restoration is $> 2.7669 \cdot 10^{-4}$, the restoration costs at most $2.1148 \cdot 10^{-4}$, and the final margin exceeds $6.52 \cdot 10^{-5}$. The analytic inputs (a new incidence Type II estimate and a new Type III estimate, proved in the companions) are Claimed; its Lean development takes 262 numerical inequalities as hypotheses. The 39-tuple printed in [16, §12] is admissible of diameter 182 (Lean-checked: `boundedGaps_182_of_DHL_39'`; the project already had another such tuple). The 726 coefficients and the source-row tables of the restoration step are not printed

in the bundle; they are in a separate repository that this project has not obtained. So no part of the variational step of 182 can be recomputed yet.

A floor for one-band hybrid supports (our own). Polymath8b's ε-simplex is the case $A_1 = 1/4$ without caps, so the following extends Theorem 15 to every one-band support of Stadlmann's type.

Theorem 45 (Proved here for the reduction; Computed, interval arithmetic, for the table)**.** *Let $T = T_k(\delta, (-\varepsilon, A_1), \underline{B}, \varepsilon)$ be a one-band support with $\varepsilon \geq 0$ and $A_1 \leq \frac{1}{4} + \omega$, with any δ and any caps. If the criterion $k(1-c_1)J(F) - kc_2K(F) > I(F)$ holds for some F supported on T, then $M_k\left(4(\frac{1}{4}+\omega+\varepsilon), 4(\frac{1}{4}+\omega-\varepsilon)\right) > 4$. This is false, for every $0 \leq \varepsilon < \frac{1}{4}+\omega$, in each of the following cases:*

ω	0	3/1000	1/300	7/1000	1/68
no k with	$k \leq 29$	$k \leq 28$	$k \leq 28$	$k \leq 26$	$k \leq 22$

So, whatever δ, the caps and the minorant, a one-band support cannot give DHL$[k, 2]$ *for $k \leq 28$ at Stadlmann's level ($\omega = 0.003$; the helper's reconstruction of his conditions gives $\omega \leq 0.0032$), for $k \leq 26$ at the level $\omega = 0.007$ of* [3]*, and for $k \leq 22$ whenever $\omega \leq 1/68$, a range containing every ω that satisfies the Type I condition $68\omega + 14\delta < 1$ of* [19, Lemma 5].

Proof. As $c_1, c_2 \geq 0$ and $K(F) \geq 0$, the criterion implies $kJ(F) > I(F)$. In the scaling $u = 4t$, F becomes a function supported on $L\mathcal{R}_k$ (because $T \subseteq \{\sum t_i \leq A_1 + \varepsilon\}$), the ratio is multiplied by 4, and J is the marginal functional over $\tau\mathcal{R}_{k-1}$; so $4 < M_k(4(A_1+\varepsilon), 4(A_1-\varepsilon))$, and monotonicity in L and τ replaces A_1 by $\frac{1}{4}+\omega$. (For $\varepsilon < 0$ one has $\tau > L$ and the quantity is at most LM_k, which is covered by the case $\varepsilon = 0$ below.) Monotonicity in k reduces each entry to the largest k shown. For the numerical part, put $A = \frac{1}{4}+\omega$. When $4(A+\varepsilon)\frac{k}{k-1}\log k < 4$ we use $M_k(L,\tau) \leq M_k(L,L) = LM_k \leq L\frac{k}{k-1}\log k$ [15, Cor. 6.4]. The remaining range of ε, up to A, is cut into intervals of length at most $1/2000$; on $[e, e']$ we take $(L,\tau) = (4(A+e'), 4(A-e))$, which dominates by monotonicity, and certify $\sup_\sigma W(\sigma) < 4$ with rational (a, s_1) exactly as in the proof of Theorem 15. The certificates are `code/cert_floor_hybrid_k*_w*.json`, written by `code/floor_hybrid.py` in ball arithmetic (`python-flint`) and re-checked by the same script with `--check` in a different interval library (`mpmath.iv`). □

Remark 46 (how sharp). For each ω in the table the same weight fails at the next k: the floating-point $\sup W$ reaches 4 on some interval of ε (for example $k = 27$, $\omega = 0.007$, near $\varepsilon \approx 0.0548$). So the table shows the reach of this weight, not of the method; the true floor is presumably much higher (Heuristic): the cap-free quantity of Proposition 22 exceeds 4 at $k = 45$ only by 0.012, so its threshold is probably not far below $k = 45$, by analogy with Remark 20. The theorem ignores the caps entirely, which only shrink the support; a floor that sees the caps is the natural next step. It also does not apply to supports with several bands, where J contains cross terms between bands, nor to the constructions behind 186 and 182, which we have not read in enough detail.

6. Programme

Track A: lowering the bound.

(1) *Climb the ladder.* The certified pipeline now stands at $k = 50$ ($H_1 \leq 246$, the established bound, via the ε-enlarged simplex). Stadlmann's region ($k = 49$, 240): version 0.6 reproduces the variational step exactly (Theorem 26, Corollary 27), so the project's bound is 240 *conditionally on the unverified reduction* [19, Prop. 1] *and equidistribution estimates of* [19]. For 212 ($k = 45$, [3]) version 0.7 confirms the variational criterion

for the caps of their Table 3 (Proposition 29); before the project's bound can move, the discrepancy between their Table 3 and Lemma 9.1 must be resolved; version 0.8 has the tool for optimising on the capped support (Proposition 31), and for the caps of Lemma 9.1 no polynomial in our 846-dimensional space reaches 3.99 (Proposition 33, resting on one program's Gram weights and a randomised check), which suggests that the draft's value belongs to its Table 3 caps. For 236 ($k = 48$) the variational criterion of [8] holds with our own degree-22 vector (Proposition 35), and the capped numerator of the published witness of [18] is reproduced exactly, integer for integer, by two independent programs (Proposition 36), so the project's bound is 236 conditionally (Corollary 37); the analytic input remains CLAIMED. Our own vector is confirmed by two programs (version 0.12); Proposition 39 makes such checks cheaper when only the inequality is needed. Then 186 and 182.

(2) *Floor of the method (our main original direction).* For the plain simplex the dual bound $M_k \le \frac{k}{k-1}\log k$ shows where the method stops ($k \ge 51$). For the ε-enlarged simplex Theorem 15 gives $k \ge 30$, which is still far from the expected truth near 50; Section 4 explains what a sharper weight must do, and Proposition 19 indicates (HEURISTIC) that it must depend on more of t than t_m and S (a Collatz–Wielandt weight $\psi_m = \int F\,dt_m/F$ built from a good positive test function F is the natural candidate). For the one-band hybrid supports of [19, 3], Theorem 45 gives the first such bound we know of ($k \ge 29$ at Stadlmann's level, $k \ge 27$ at that of [3]), still far from 49 and 45; it ignores the caps. A rigorous dual certificate there would say how far one-band supports can go with the present inputs, and which input (which Type I/II/III range) is binding; the constructions behind 186 and 182 are of a different kind and would need their own dual bound.

(3) *Which rung is cheap.* From Table 1, the gain per unit of k below the claimed frontier is $182 \to 176 \to 168 \to 162 \to 158$ for $k = 39, \dots, 35$: the step $38 \to 37$ (gain 8) is worth the most; $39 \to 38$ and $37 \to 36$ give 6 each.

Problem 47 (OPEN; partly answered by Theorem 45 for one band, without using the caps)**.** Give a rigorous upper bound for the variational quantity of [19] as a function of k and of the admissible exponents, and deduce the least k the hybrid method can reach with the currently proved estimates.

Track B: the parity barrier. No sieve of the above kind can distinguish integers with an even number of prime factors from those with an odd number. Known ways around the barrier all inject bilinear or structural information. We will study, one per session, what each would need in order to say something about twin primes: Chen's theorem [4]; Bombieri's asymptotic sieve [2]; the parity-breaking sieve of Friedlander and Iwaniec [6]; Heath-Brown's theorem that Siegel zeros would imply the conjecture [7]; the function-field proof of Sawin and Shusterman [17]; and the entropy-based results on correlations of multiplicative functions [20].

Problem 48 (OPEN, to be made precise in Lean)**.** State, as a single explicit hypothesis on bilinear sums over $n(n+2)$, the weakest known input that implies the twin prime conjecture, and formalise the implication.

Rules of this paper. A bound is listed as reproduced by this project only if its variational step has an exact certificate (for the 2026 bounds, confirmed by two independent programs) and its combinatorial step is Lean-checked; every unverified analytic input is named next to it. Ideas are logged with their outcome, including failures. Nothing here is peer reviewed.

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